

Recap

Part 1

- Representing circuits/Reducing circuits
 - Boolean Algebra
 - Karnaugh Maps (*K-maps*)

Part 2

- Combinational Logic/ Combinational Circuits
 - Half adder
 - Full adder
 - Half subtractor
 - Full subtractor
 - Universal half adder and full adder
 - Universal half subtractor and full subtractor
 - 4-bit binary adder
 - 4-bit binary subtractor
 - Carry look ahead adder

Part 2

- Combinational Logic/ Combinational Circuits
 - BCD adder
 - Binary multiplier
 - Magnitude Comparator
 - Multiplexer
 - Encoders
 - Decoders
 - Implementation of Boolean functions using multiplexers and decoders

Assignment 1 & 2

- Note: All circuits should be drawn neatly by hand. All the components in terms of gates and wires should be drawn, do not use symbolic diagram for any of the parts
 1. 16-bit binary adder
 2. Binary multiplier to multiply 02 8-bit numbers
 3. A circuit performing binary division
 4. 8-line priority encoder
 5. 16: 1 multiplexor
 6. 4:16 decoder

Project

- A digital electronic project that employs
 - Arduino or Raspberry pi
 - Must use a few sensors (e.g. temperature, pressure, humidity sensors)
 - Doing Something useful
 - Should be fairly complex

OR

- A digital electronic project that employs
 - Some hardware descriptive language like Verilog or VHDL
 - You should be designing some complex module e.g. a RAM, Memory, Round Robin Bus Arbiter etc.
- Note:
 - Group of 02 or 03 students
 - Arduino board is quite cheaper than Raspberry Pi module

Part 3

- Sequential logic/Sequential circuits
 - Output dependent upon present input and previous output
 - Memory element required

SEQUENTIAL LOGIC

Block Diagram of a Sequential Circuit

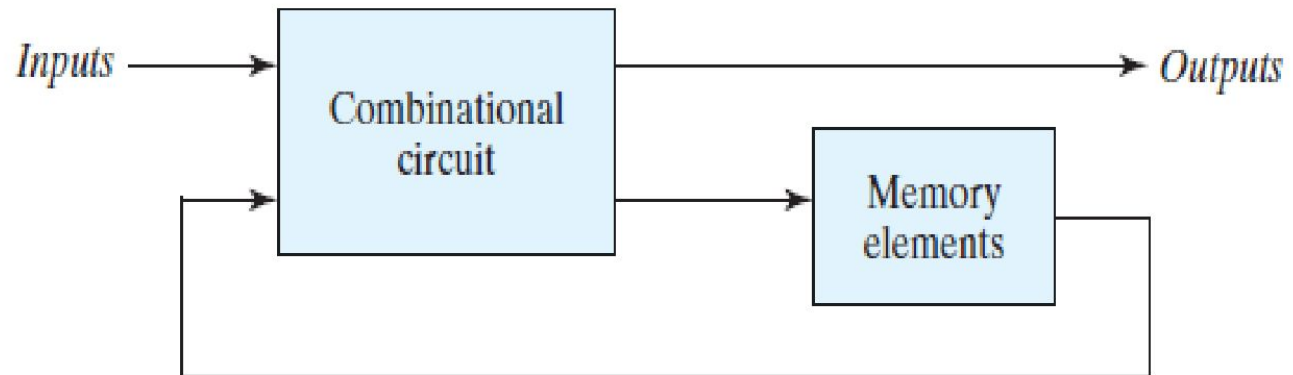
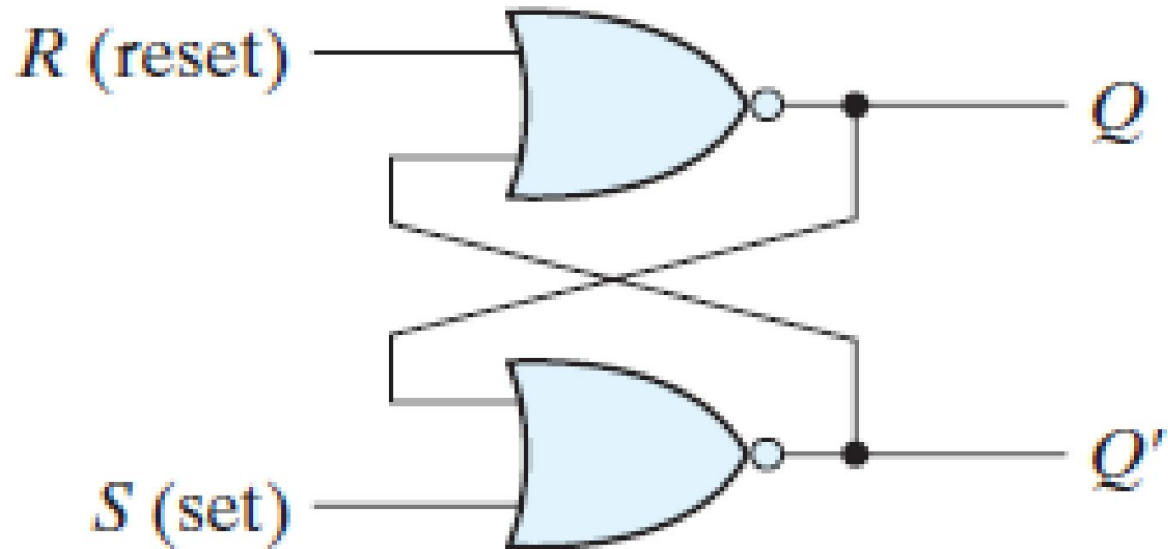


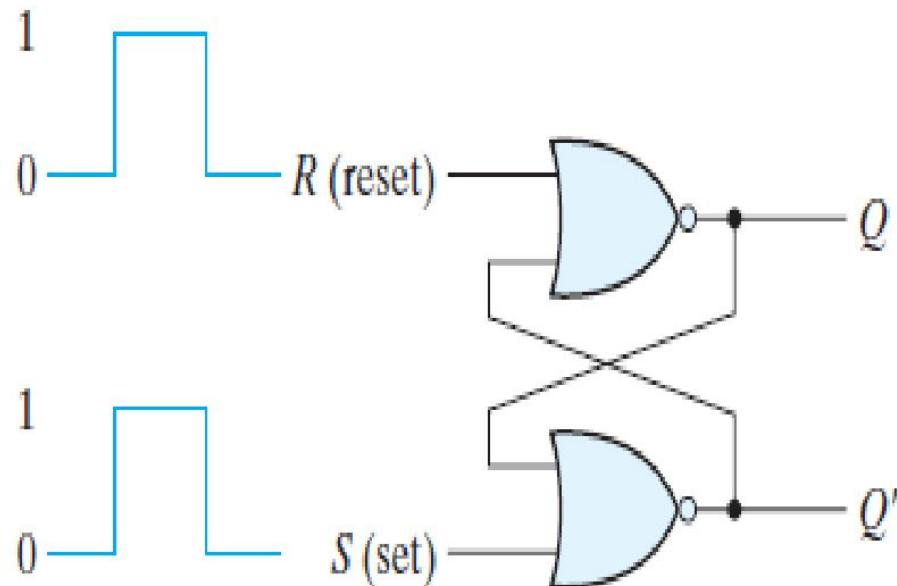
FIGURE 5.1

Block diagram of sequential circuit

SR LATCH (NOR GATE)



SR LATCH (NOR GATE)

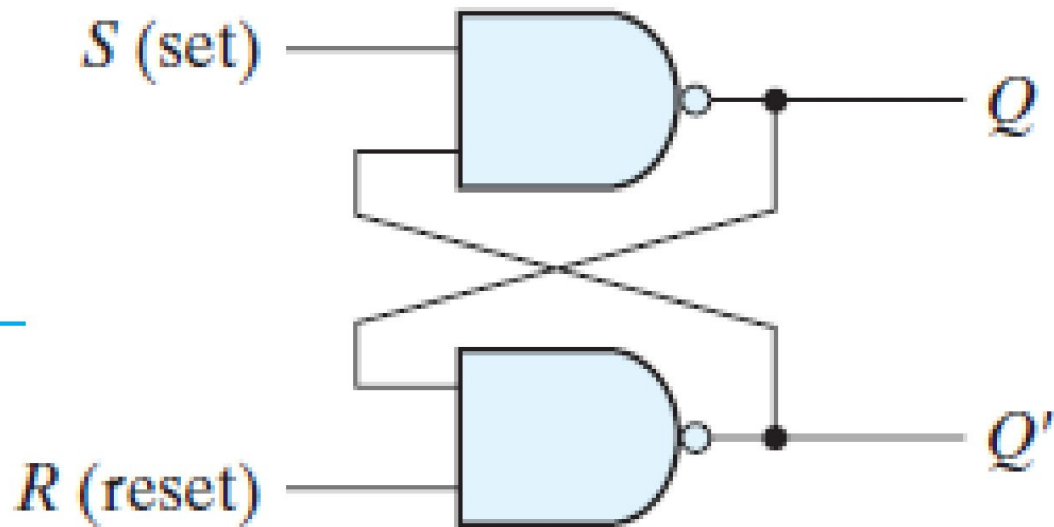


(a) Logic diagram

S	R	Q	Q'
1	0	1	0
0	0	1	0 (after $S = 1, R = 0$)
0	1	0	1
0	0	0	1 (after $S = 0, R = 1$)
1	1	0	0 (forbidden)

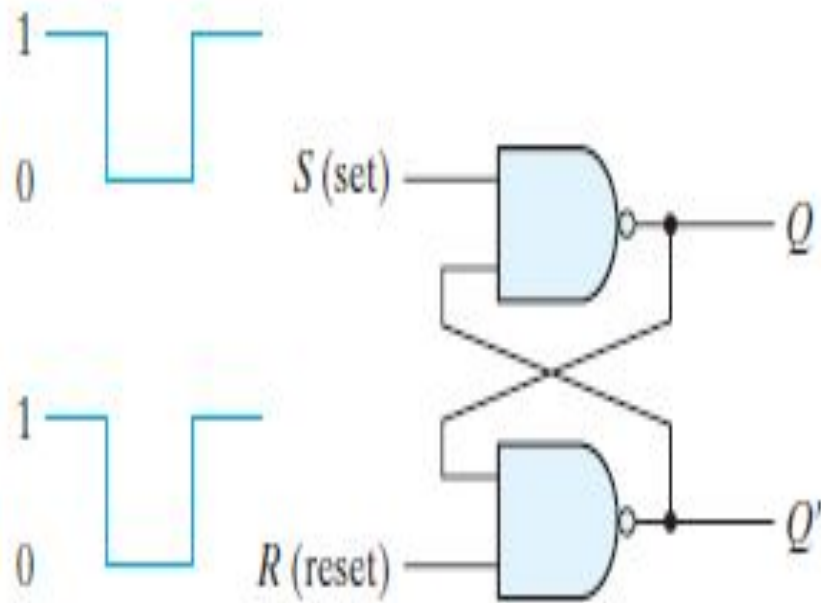
(b) Function table

SR LATCH (NAND GATE)



(a) Logic diagram

SR LATCH (NAND GATE)

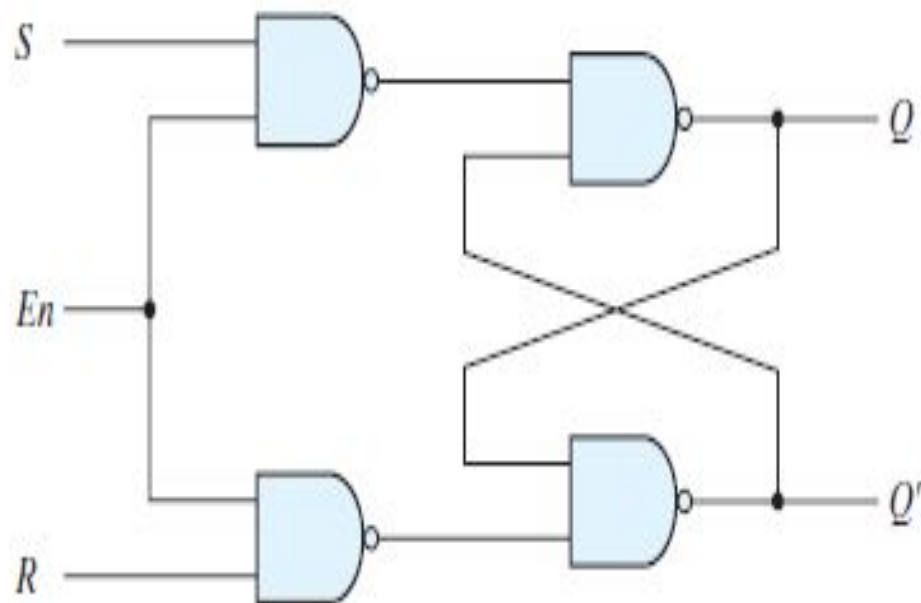


(a) Logic diagram

S	R	Q	Q'
1	0	0	1
1	1	0	1 (after $S = 1, R = 0$)
0	1	1	0
1	1	1	0 (after $S = 0, R = 1$)
0	0	1	1 (forbidden)

(b) Function table

SR LATCH with CONTROL INPUT



(a) Logic diagram

En	S	R	Next state of Q
0	X	X	No change
1	0	0	No change
1	0	1	$Q = 0$; reset state
1	1	0	$Q = 1$; set state
1	1	1	Indeterminate

(b) Function table

FIGURE 5.5

SR latch with control input

Circuit Triggering

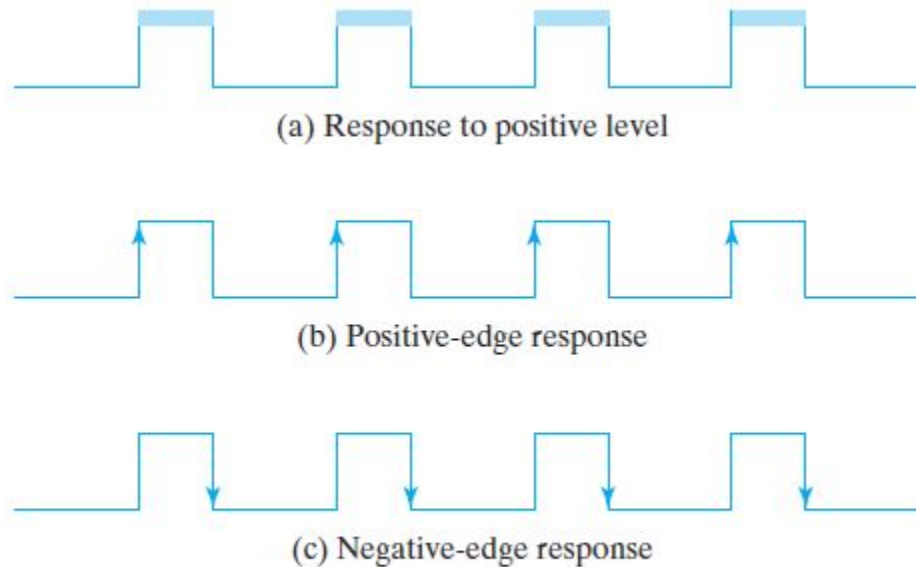
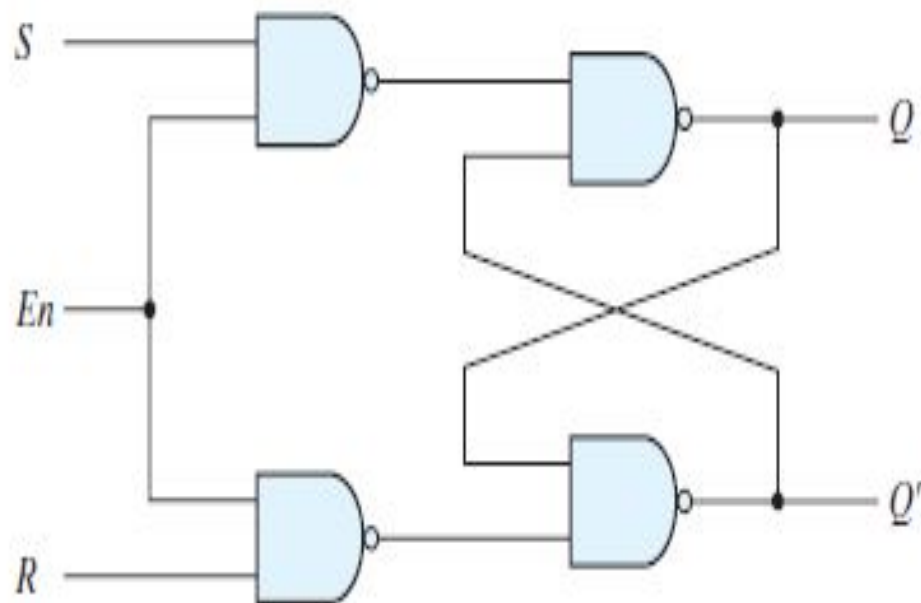


FIGURE 5.8

Clock response in latch and flip-flop

SR LATCH with CLOCK



(a) Logic diagram

En	S	R	Next state of Q
0	X	X	No change
1	0	0	No change
1	0	1	$Q = 0$; reset state
1	1	0	$Q = 1$; set state
1	1	1	Indeterminate

(b) Function table

FIGURE 5.5

SR latch with control input

SUMMARY

CLK	S	R	Q	Q'
0	X	X	Memory	Memory
1	0	0	Memory	Memory
1	0	1	0	1
1	1	0	1	0
1	1	1	Not Used	Not Used

Truth Table for SR Flip Flop

CLK	S	R	Q_{n+1}
0			
1			
1			
1			
1			

Truth Table for SR Flip Flop

CLK	S	R	Q_{n+1}
0	X	X	Q_n next state would be same as the present
1	0	0	Q_n (next state would be the same as the present)
1	0	1	0
1	1	0	1
1	1	1	Invalid

Characteristic Table for SR Flip Flop

[illegible]

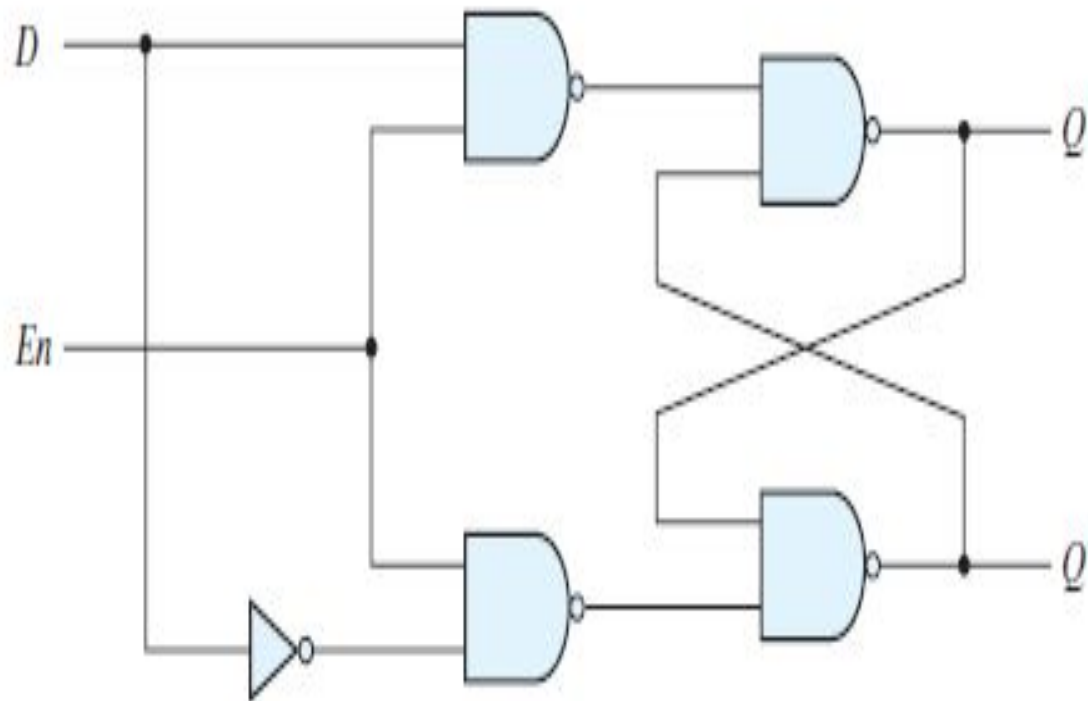
Characteristic Table for SR Flip Flop

Q_n	S	R	Q_{n+1}
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	invalid
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	invalid

Excitation Table for SR Flip Flop

Q_n	Q_{n+1}	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

D-FLIP FLOP



(a) Logic diagram

En	D	Next state of Q
0	X	No change
1	0	$\underline{Q} = 0$; reset state
1	1	$\underline{Q} = 1$; set state

(b) Function table

Truth Table for D Flip Flop

CLK	D	Q_{n+1}
0	X	Q_n next state would be same as the present
1	0	0
1	1	1

Characteristic Table for D Flip Flop

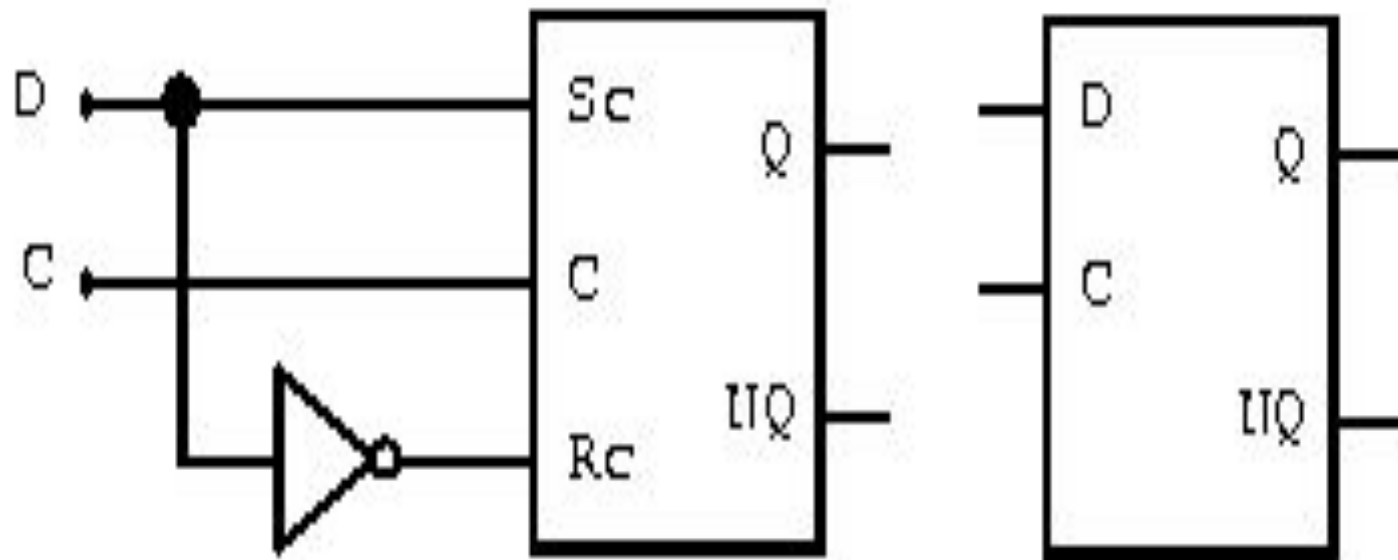
Q_n	D	Q_{n+1}
0	0	0
0	1	1
1	0	0
1	1	1

Excitation Table for D Flip Flop

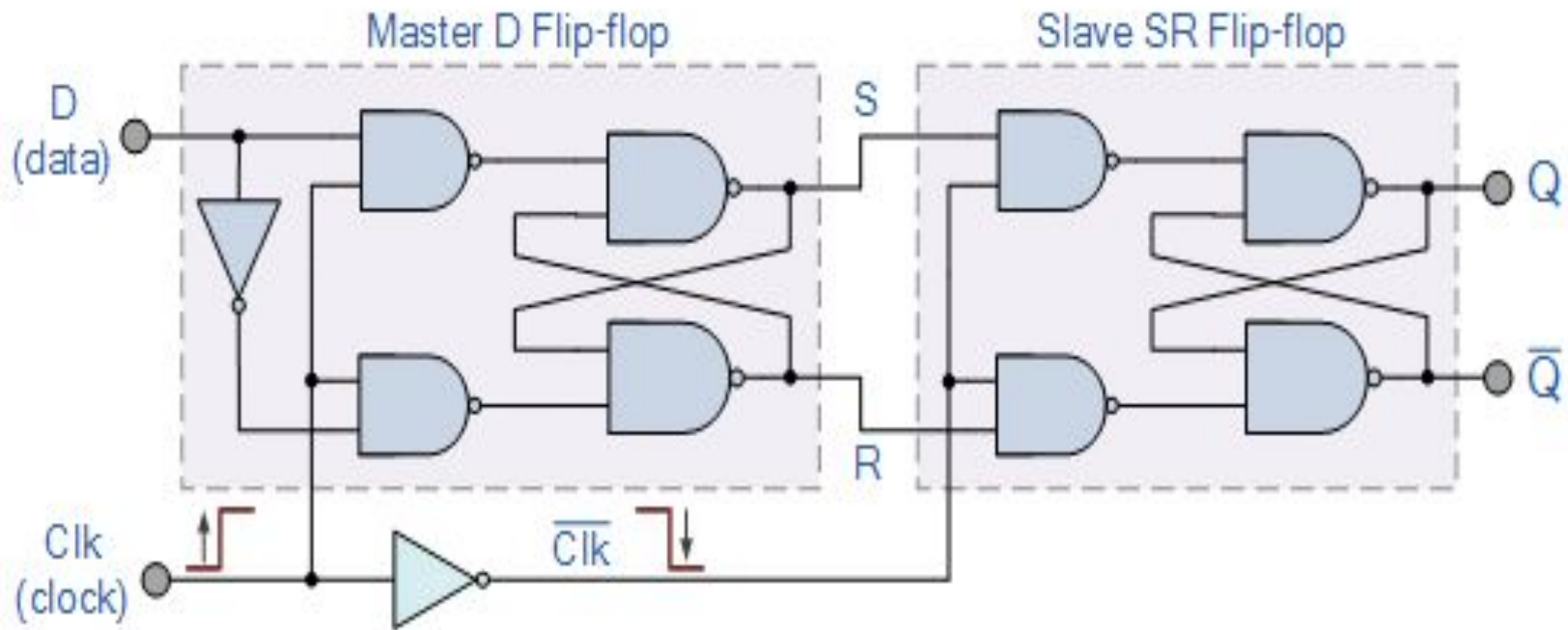
Q_n	Q_{n+1}	D
0	0	0
0	1	1
1	0	0
1	1	1

D-FLIP FLOP

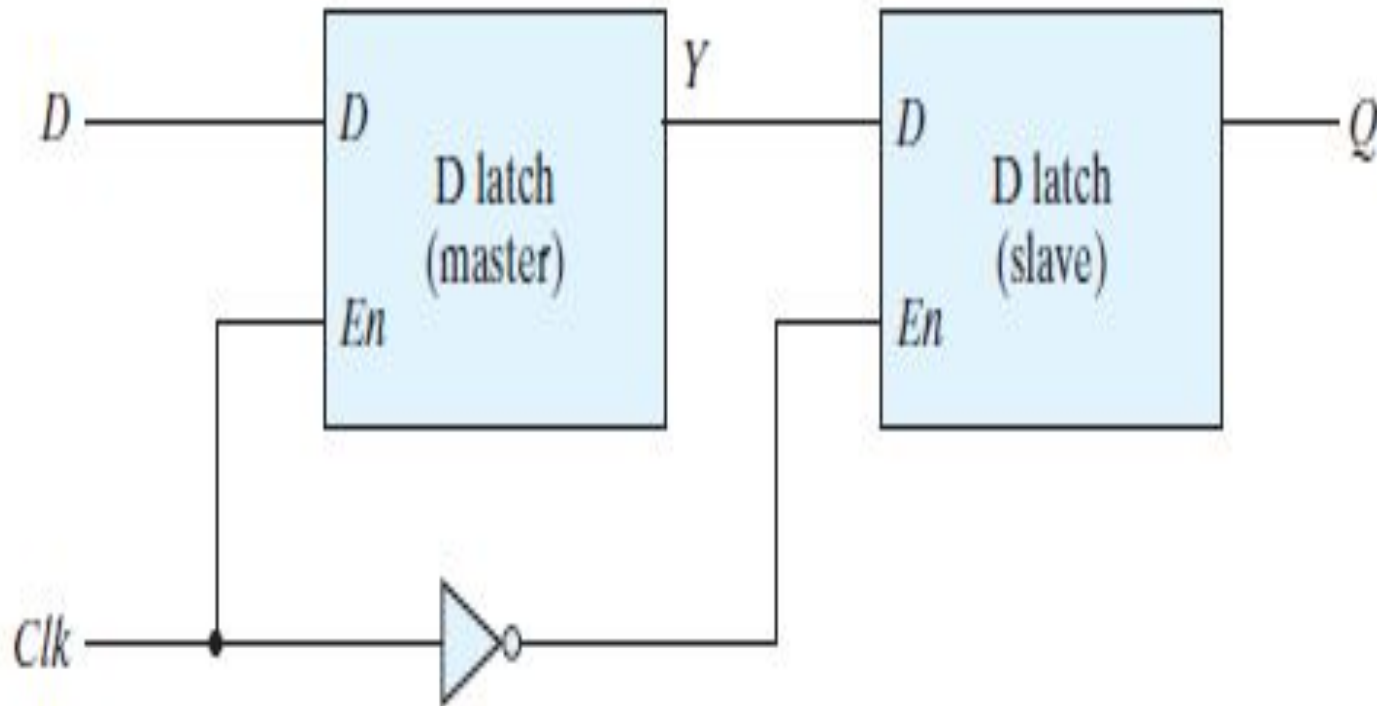
D FLIP FLOP



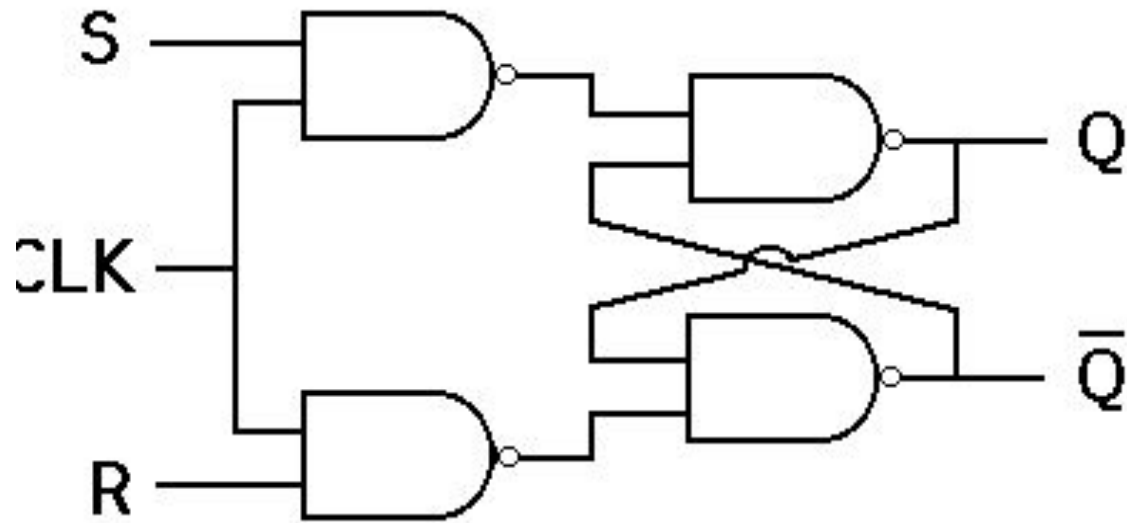
Master Slave D-Flip Flop



Master Slave D-Flip Flop



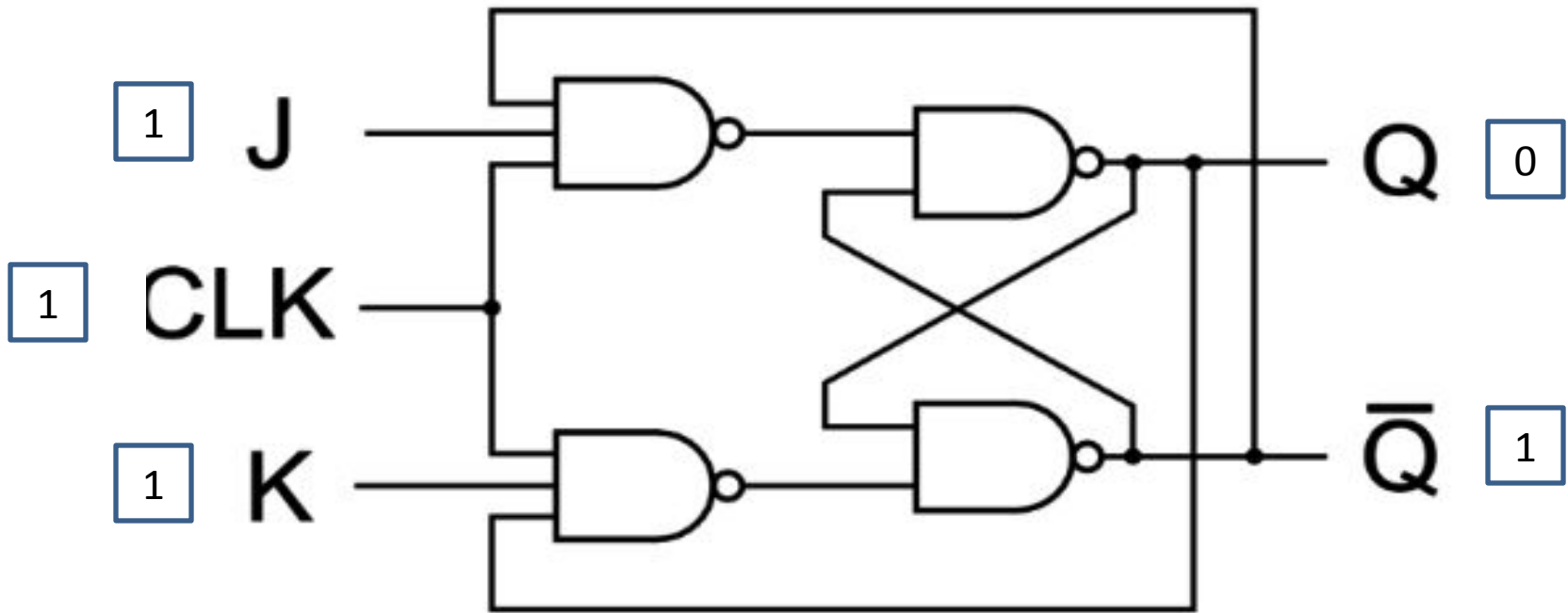
SR Flip Flop



Truth Table for D Flip Flop

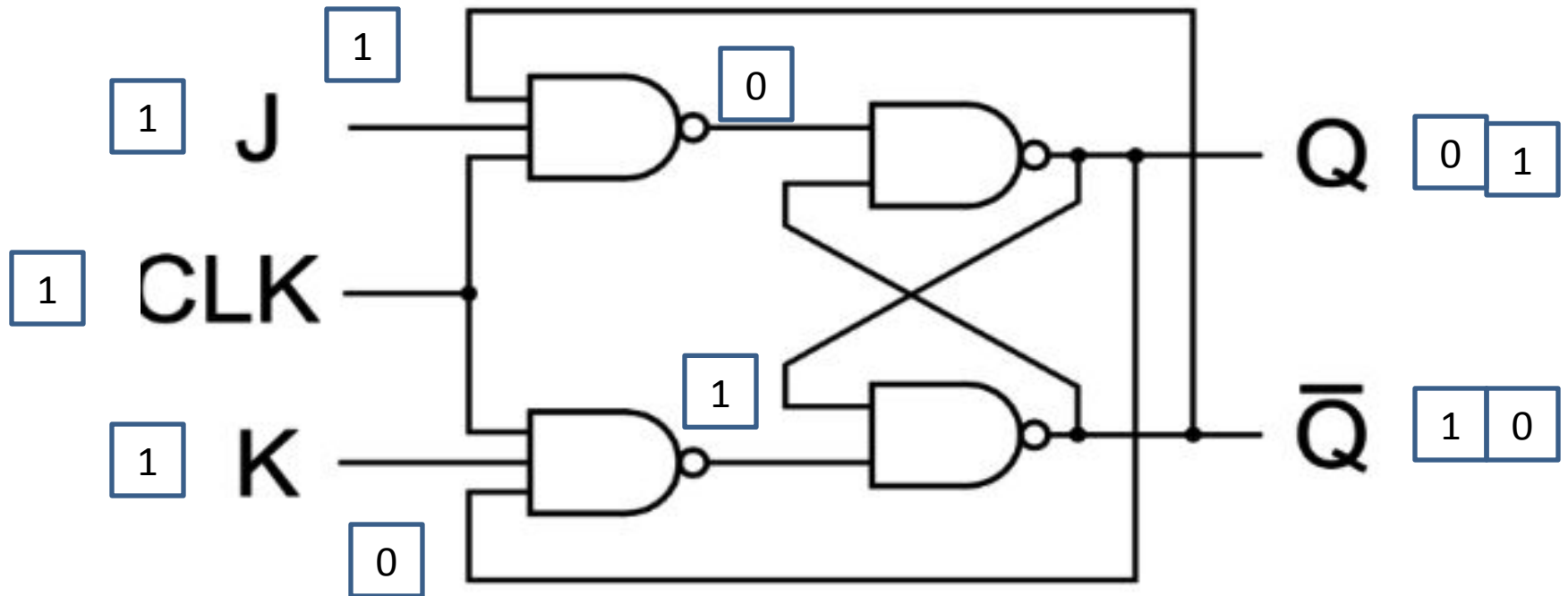
CLK	S	R	Q_{n+1}
0	X	X	Q_n next state would be same as the present
1	0	0	Q_n (next state would be the same as the present)
1	0	1	0
1	1	0	1
1	1	1	Invalid

JK Flip Flop



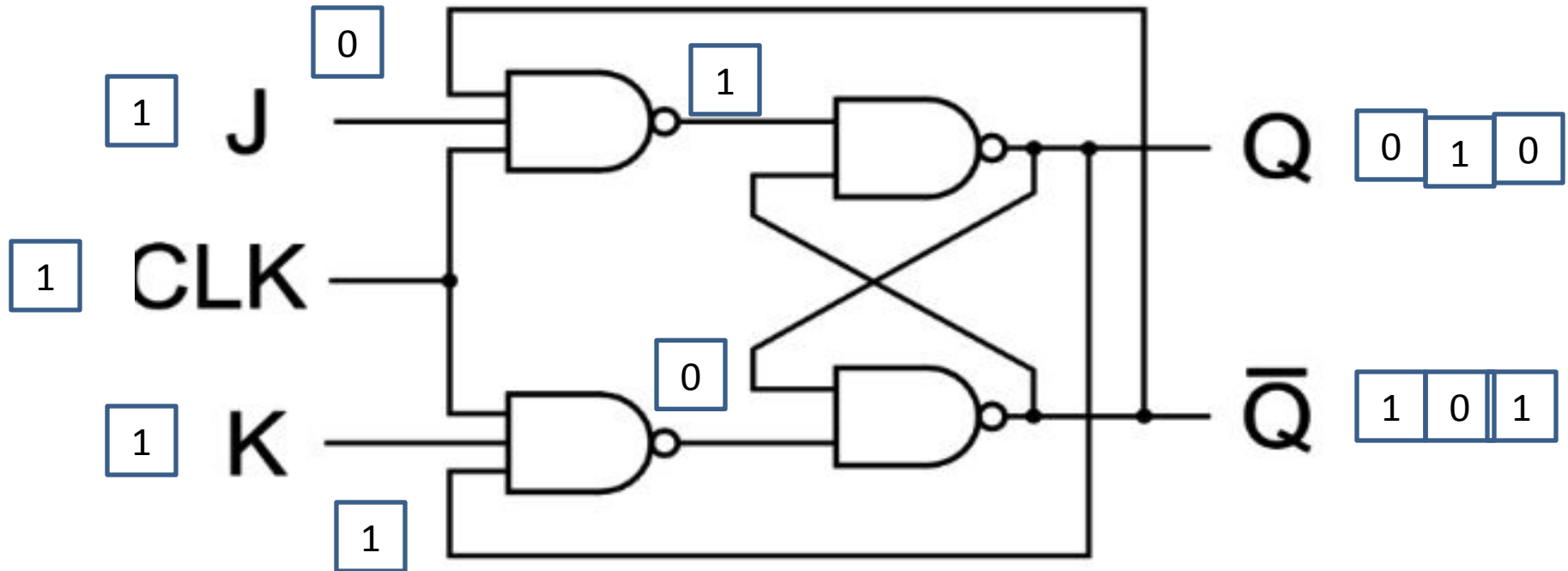
S	R	Q	Q'
0	0	Not used	
0	1	1	0
1	0	0	1
1	1	Memory	

JK Flip Flop



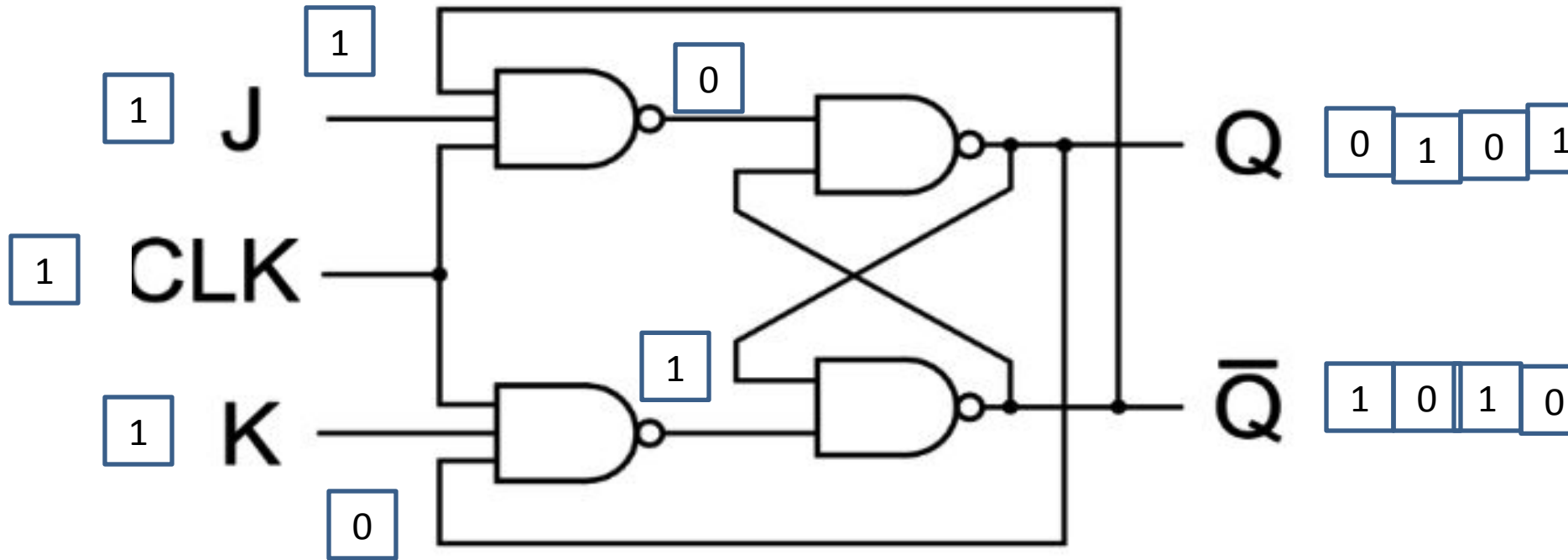
S	R	Q	Q'
0	0	Not used	
0	1	1	0
1	0	0	1
1	1	Memory	

JK Flip Flop



S	R	Q	Q'
0	0	Not used	
0	1	1	0
1	0	0	1
1	1	Memory	

JK Flip Flop



S	R	Q	Q'
0	0	Not used	
0	1	1	0
1	0	0	1
1	1	Memory	

Truth Table for JK Flip Flop

CLK	J	K	Q_{n+1}
0	X	X	Q_n
1	0	0	Q_n
1	0	1	0
1	1	0	1
1	1	1	Q'_n

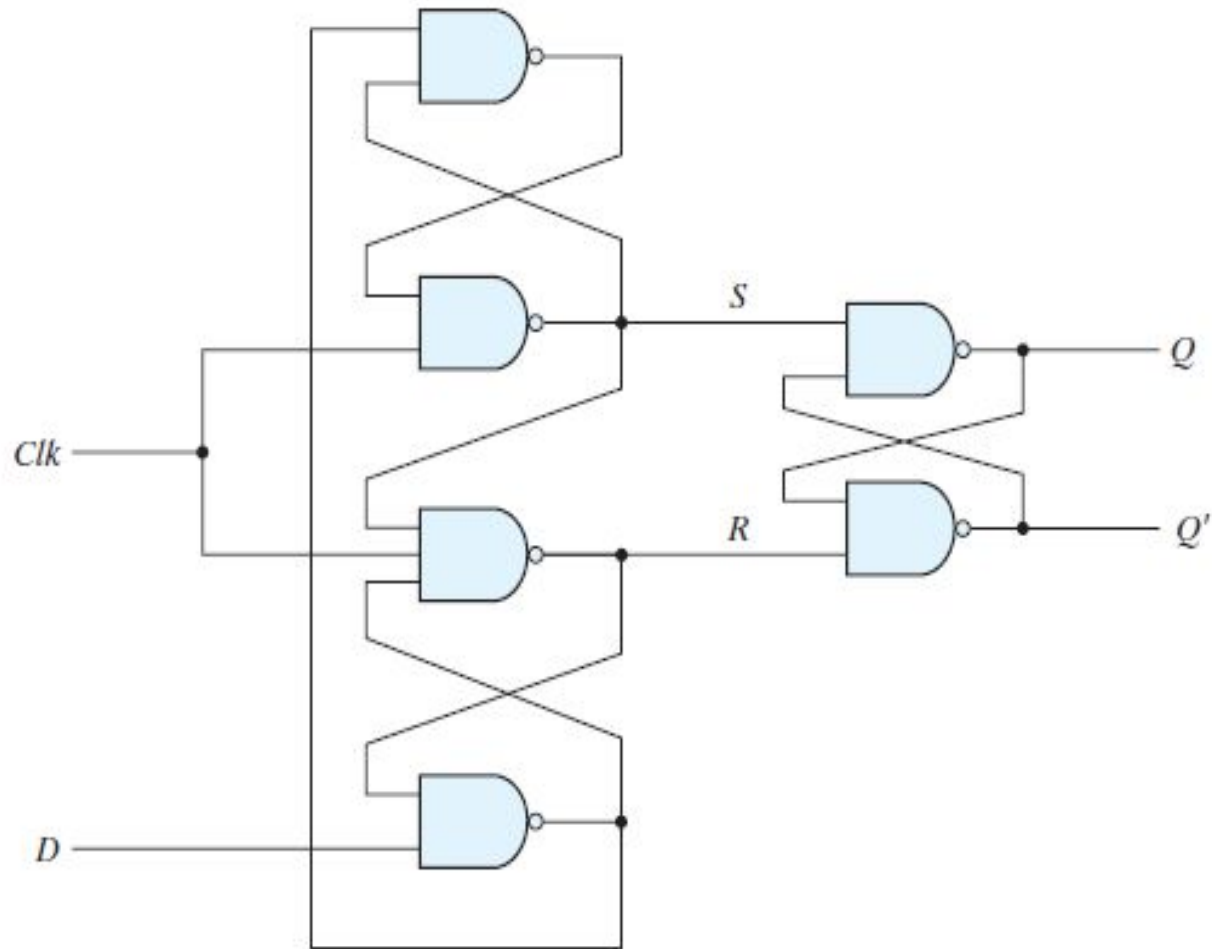
Characteristic Table for JK Flip Flop

Q_n	J	K	Q_{n+1}
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

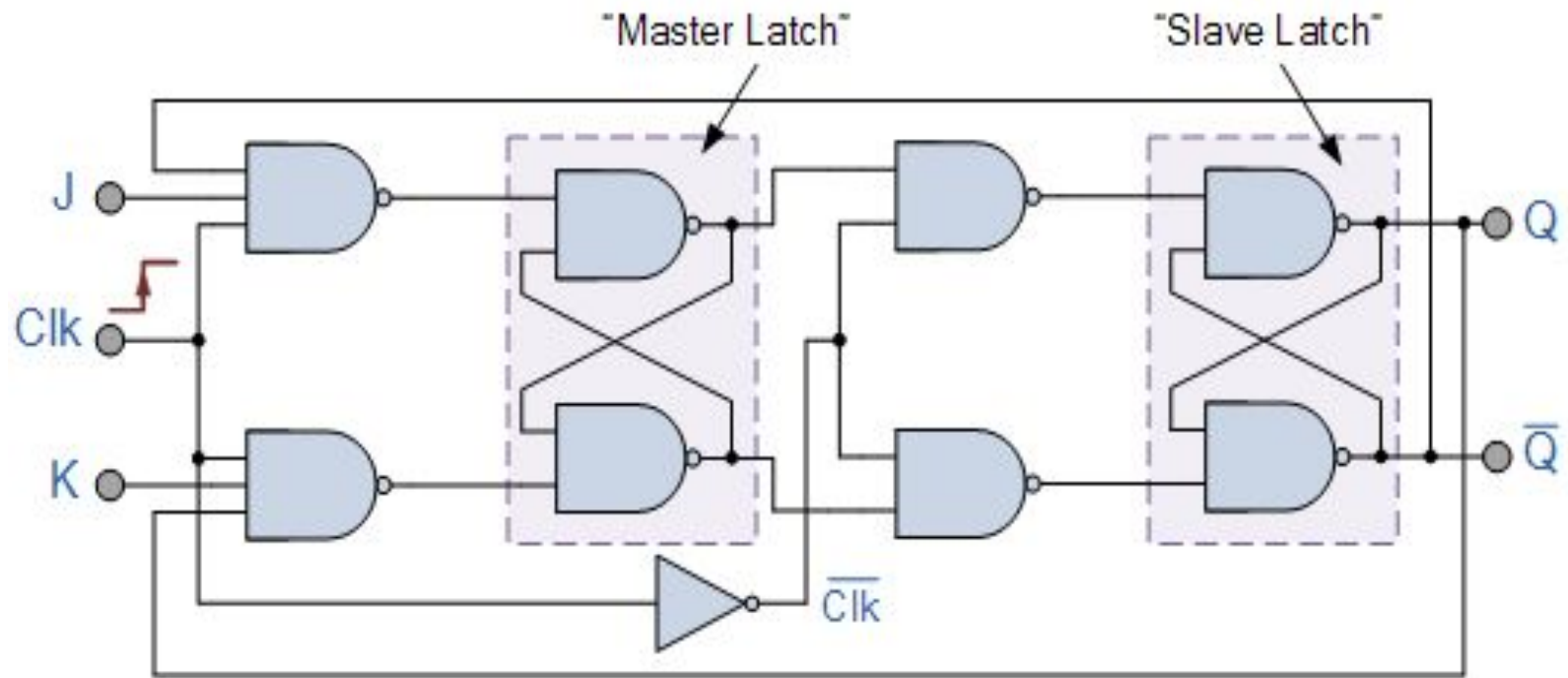
Excitation Table for SR Flip Flop

Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

D Type Positive Edge Triggered Flip Flop

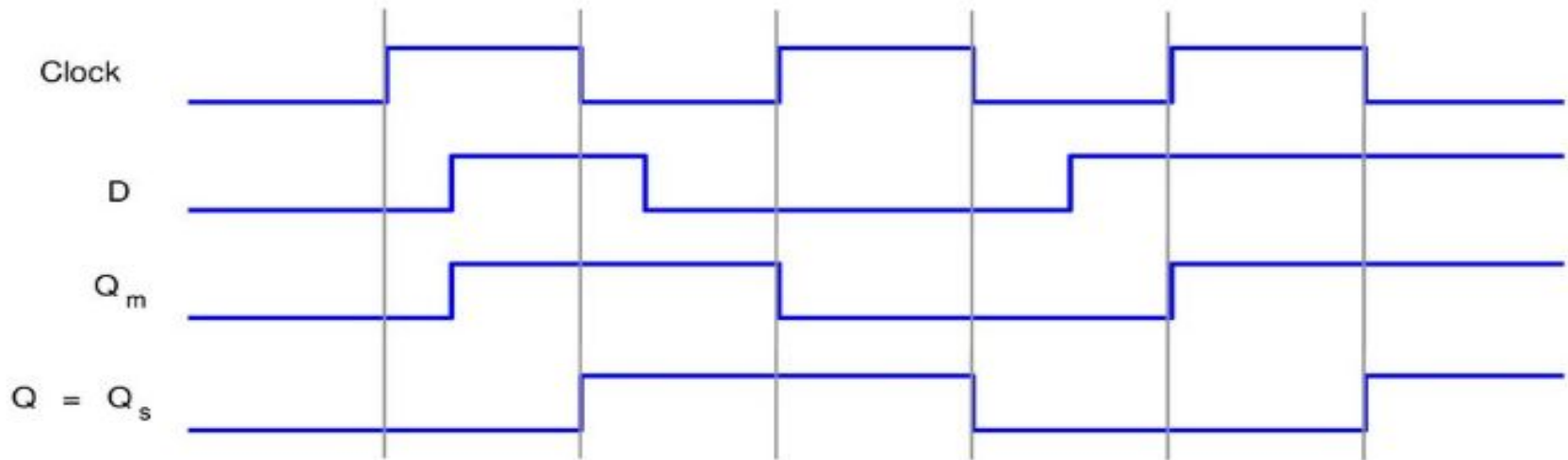
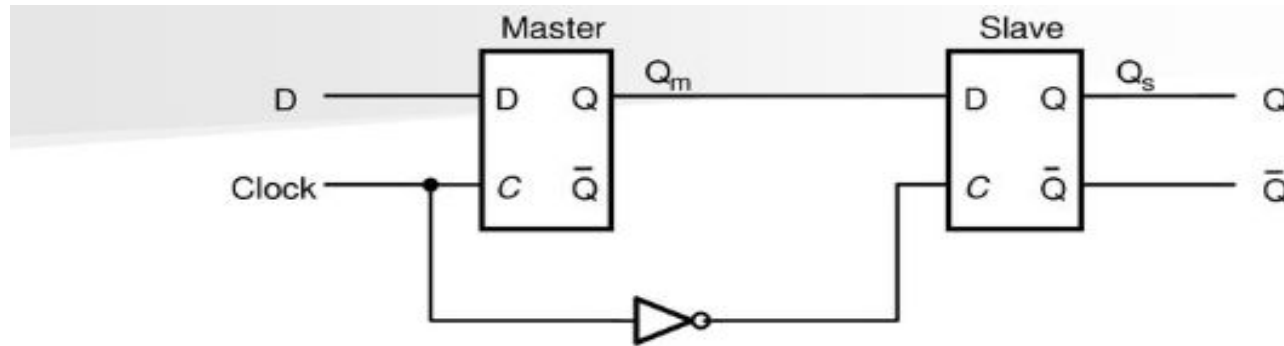


Master Slave JK Flip Flop



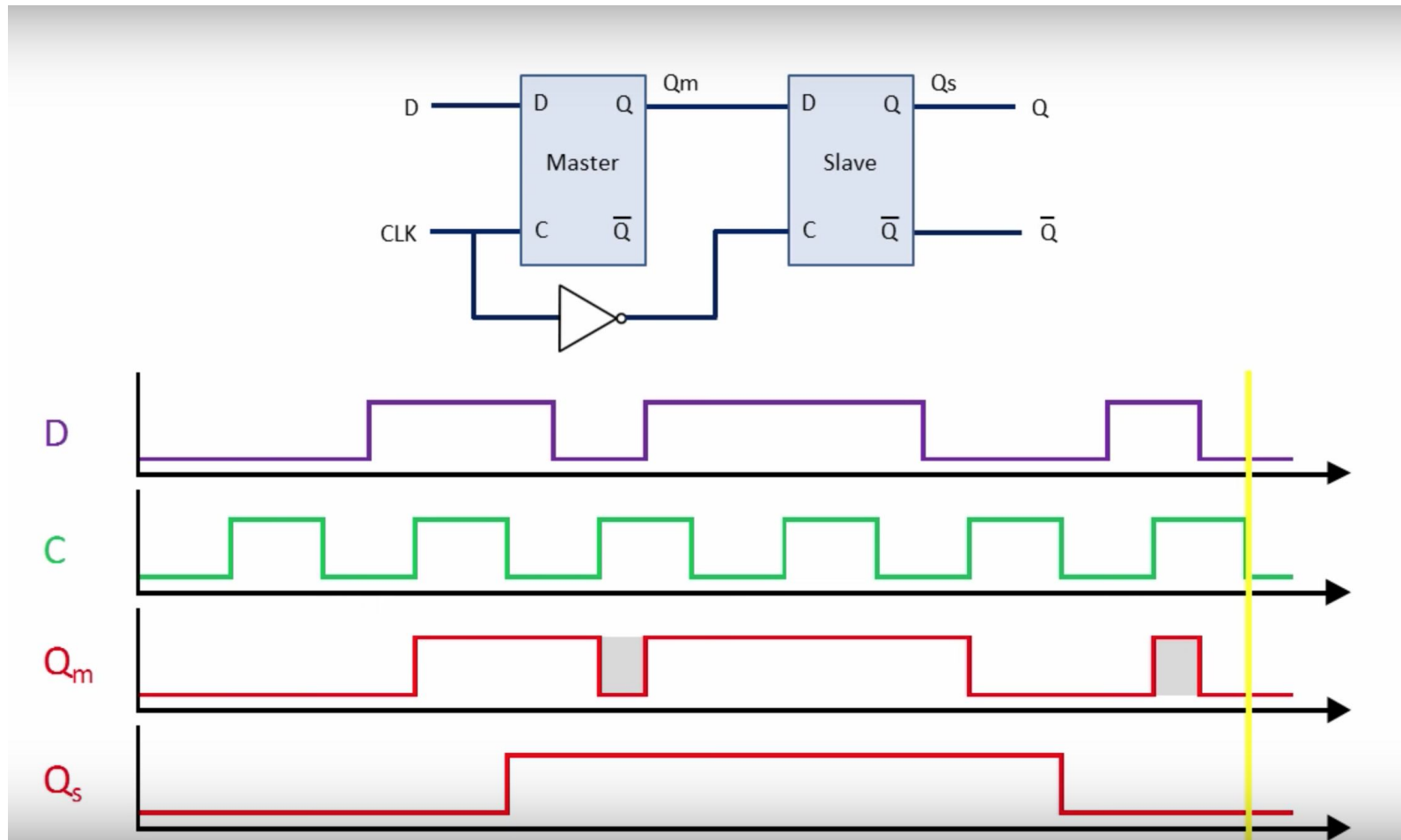
Timing Diagram for Master Slave JK Flip Flop

Timing Diagram for Master Slave D Flip Flop

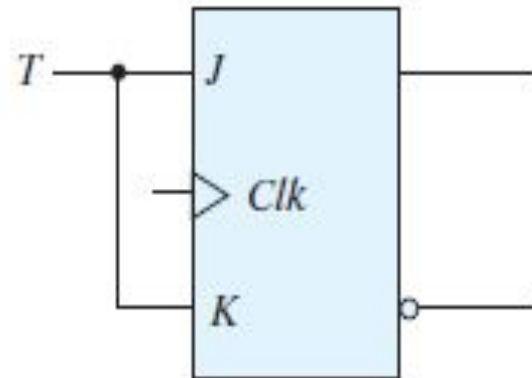
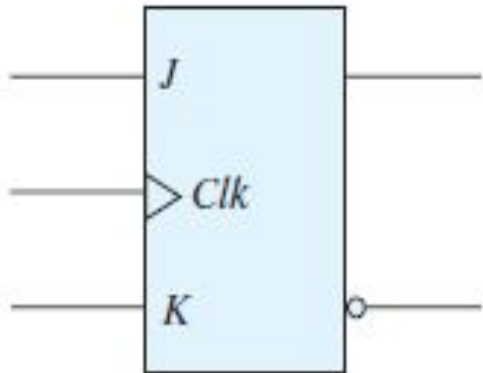


Timing diagram

Timing Diagram for Master Slave D Flip Flop

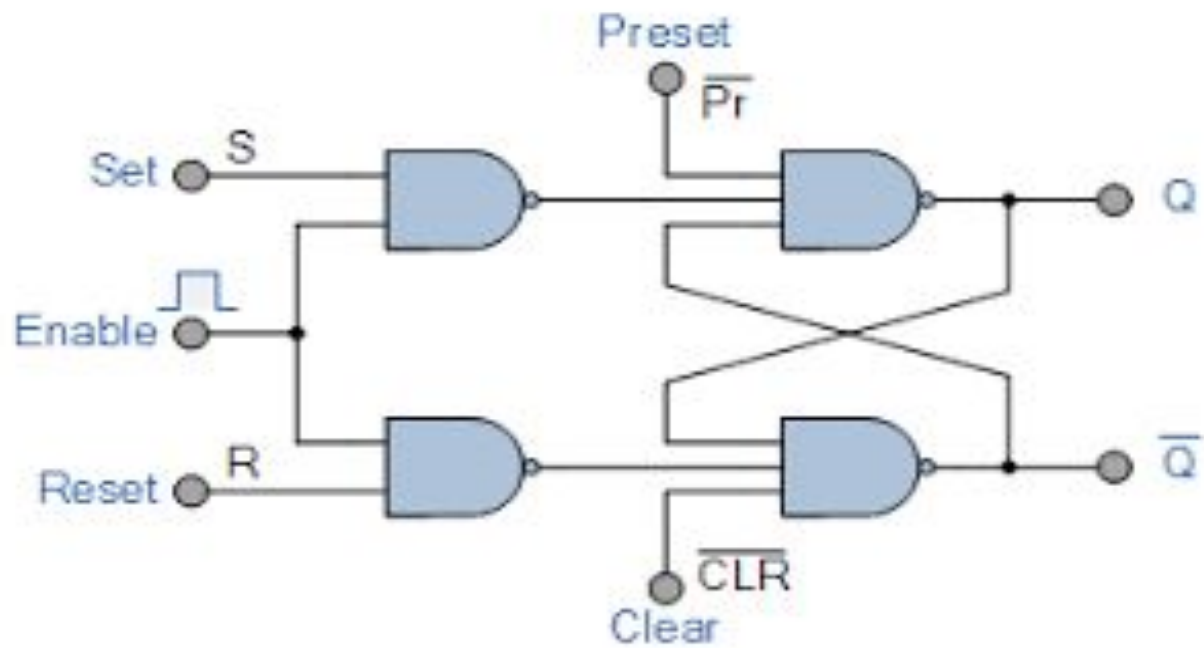


T Flip Flop



Tables

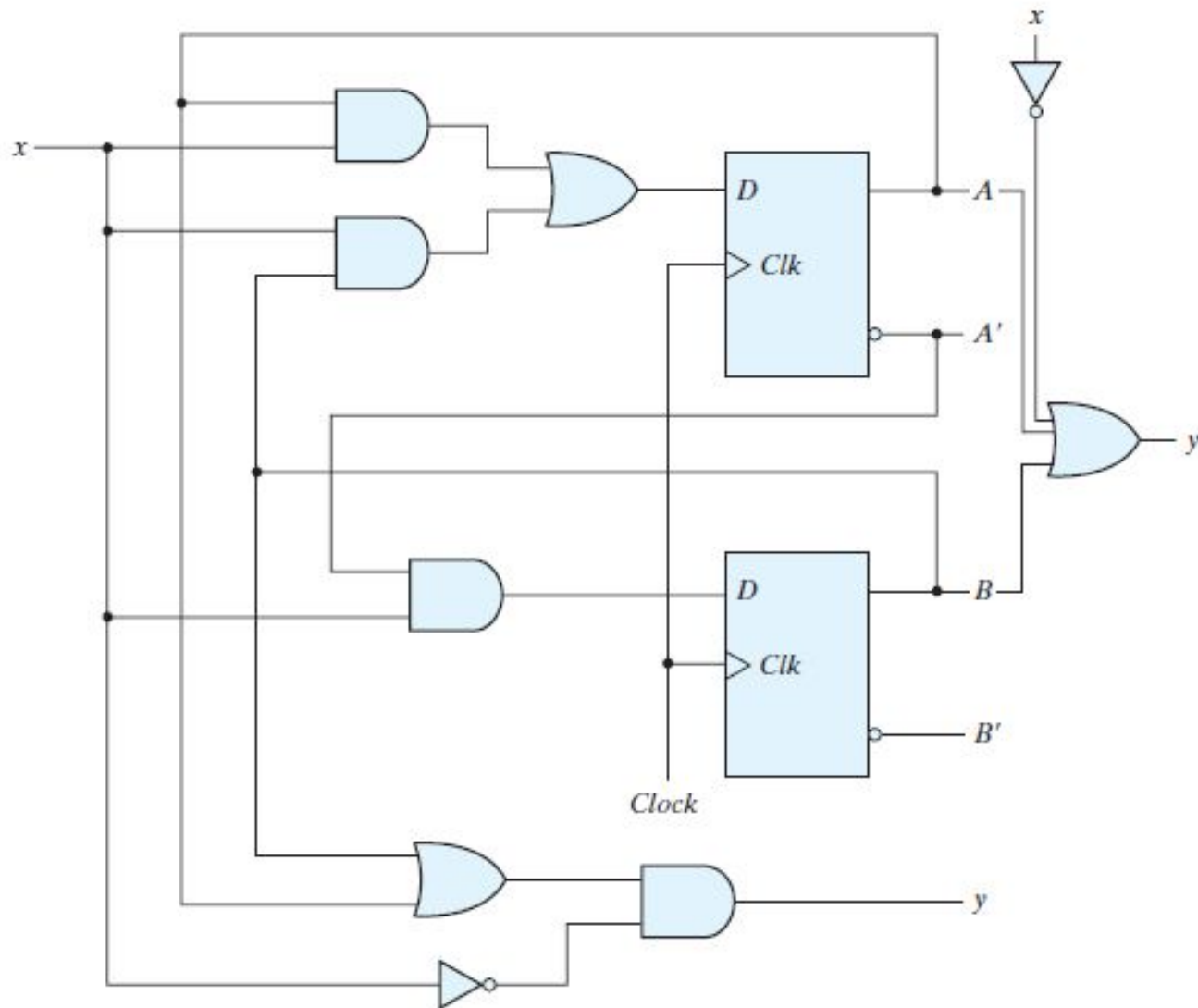
Direct Inputs



Direct Inputs

Preset	Clear	Qn
0	0	Not used
0	1	1
1	0	0
1	1	Flip Flop Behaves normally

Analysis of Clocked Sequential Circuits



Analysis of Clocked Sequential Circuits

Table 5.2
State Table for the Circuit of Fig. 5.15

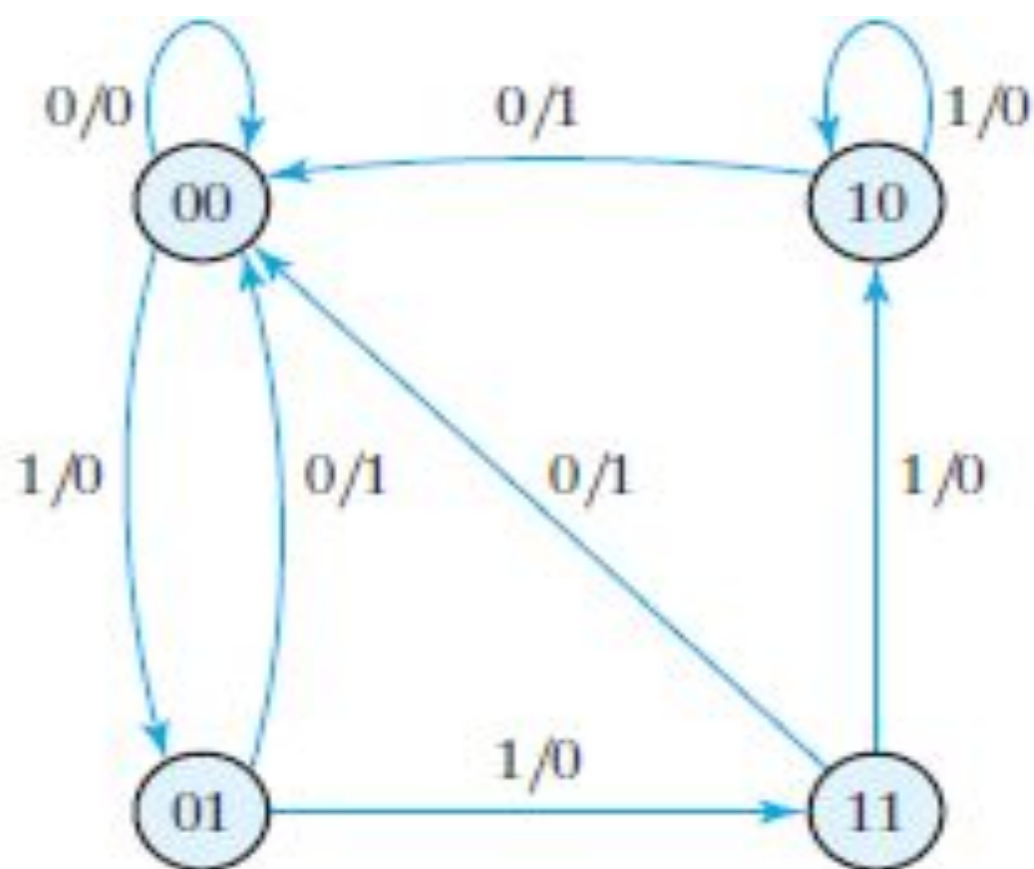
Present State		Input	Next State		Output
A	B		A	B	
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	0	1
0	1	1	1	1	0
1	0	0	0	0	1
1	0	1	1	0	0
1	1	0	0	0	1
1	1	1	1	0	0

Analysis of Clocked Sequential Circuits

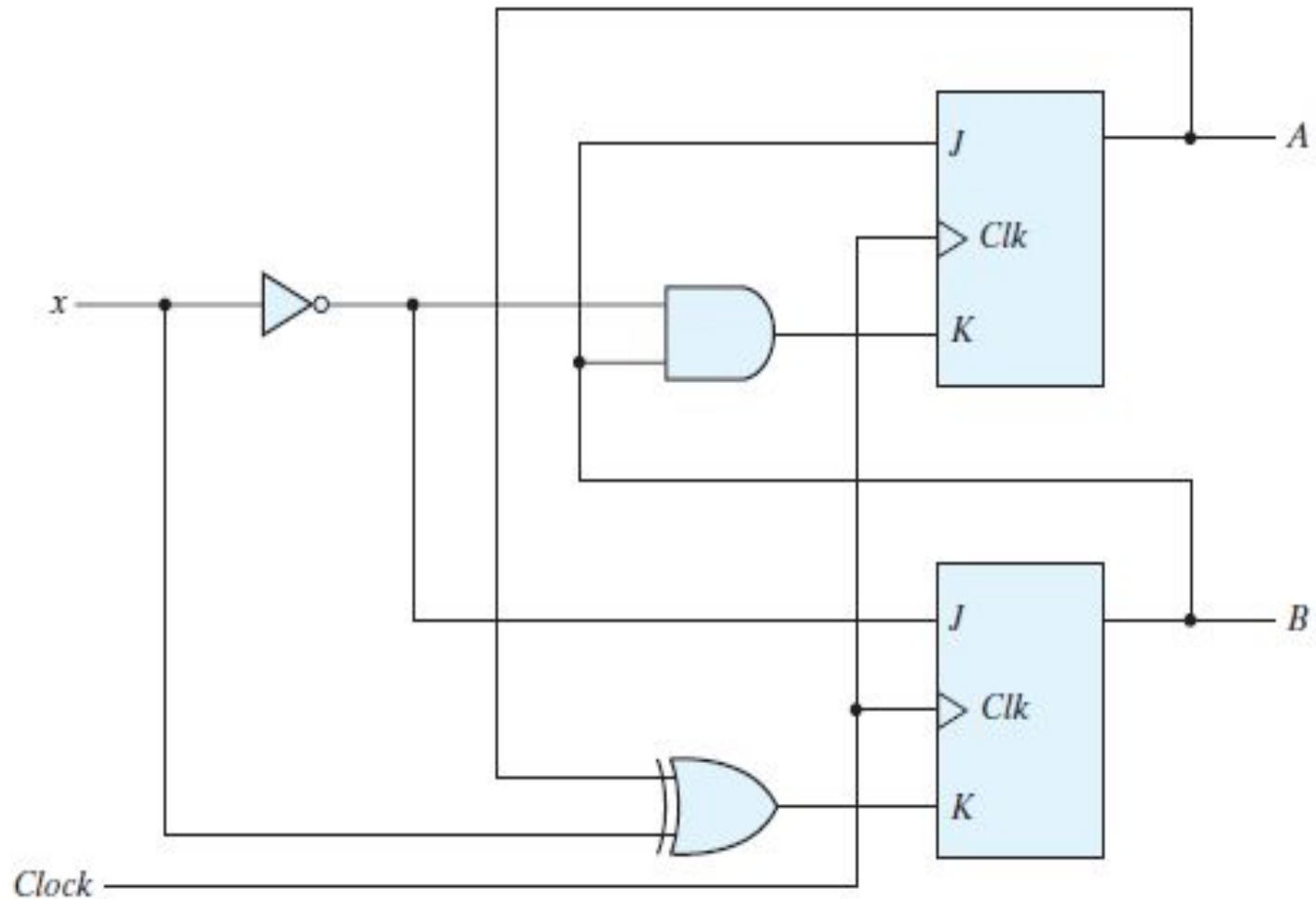
Second Form of the State Table

Present State		Next State				Output	
		$x = 0$		$x = 1$		$x = 0$	$x = 1$
<i>A</i>	<i>B</i>	<i>A</i>	<i>B</i>	<i>A</i>	<i>B</i>	<i>y</i>	<i>y</i>
0	0	0	0	0	1	0	0
0	1	0	0	1	1	1	0
1	0	0	0	1	0	1	0
1	1	0	0	1	0	1	0

Analysis of Clocked Sequential Circuits



Analysis with JK Flip Flop

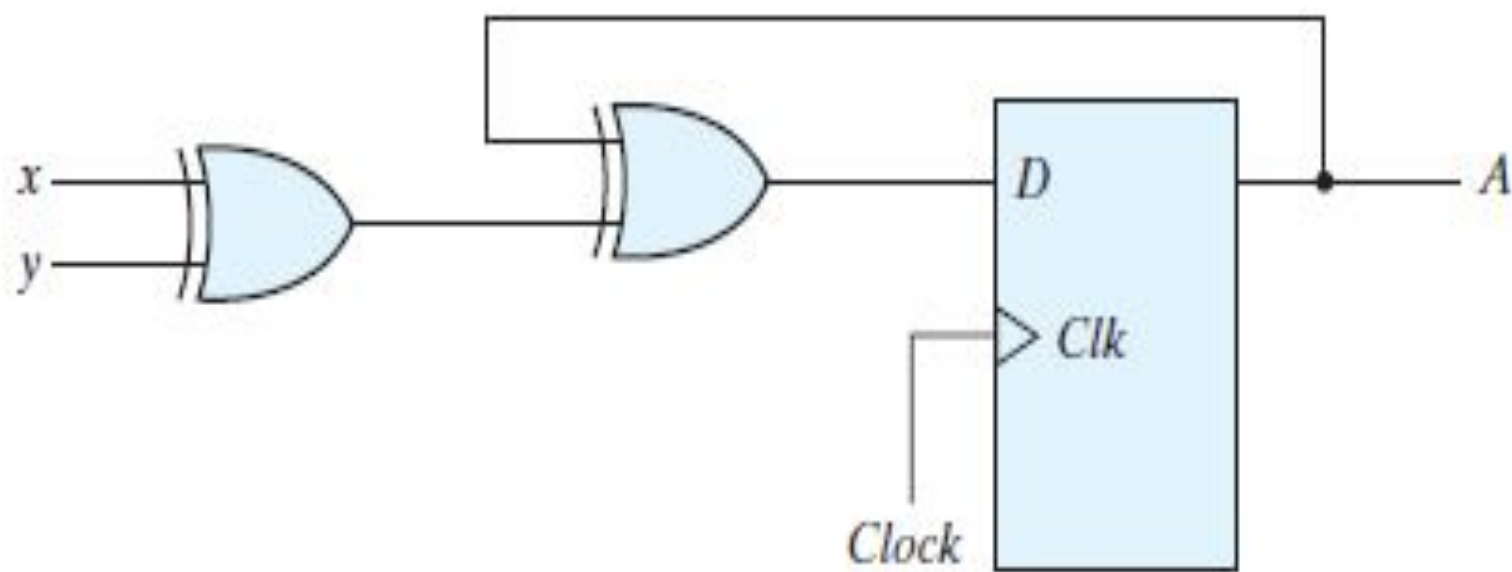


Analysis with JK Flip Flop

Table 5.4
State Table for Sequential Circuit with JK Flip-Flops

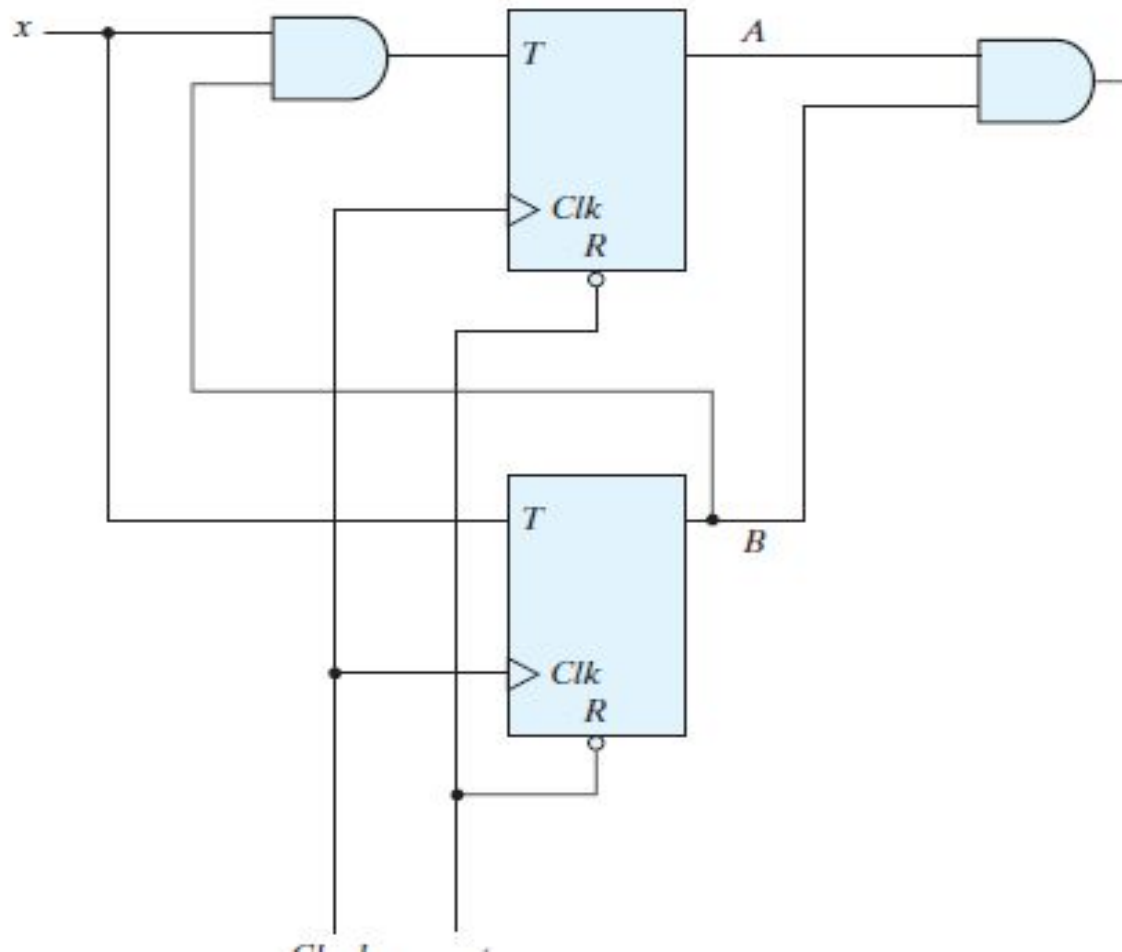
Present State		Input	Next State		Flip-Flop Inputs			
A	B	x	A	B	J_A	K_A	J_B	K_B
0	0	0	0	1	0	0	1	0
0	0	1	0	0	0	0	0	1
0	1	0	1	1	1	1	1	0
0	1	1	1	0	1	0	0	1
1	0	0	1	1	0	0	1	1
1	0	1	1	0	0	0	0	0
1	1	0	0	0	1	1	1	1
1	1	1	1	1	1	0	0	0

Analysis with JK Flip Flop



(a) Circuit diagram

Analysis with T Flip Flop



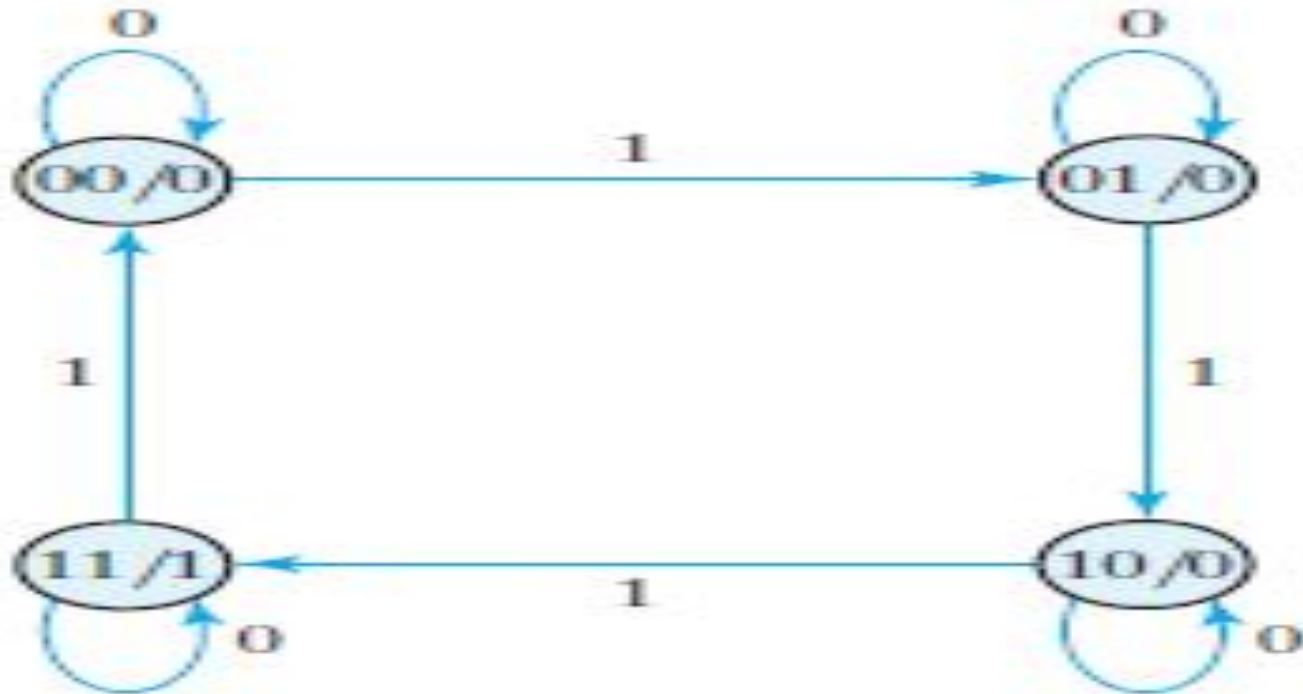
Analysis with T Flip Flop

Table 5.5

State Table for Sequential Circuit with T Flip-Flops

Present State		Input	Next State		Output
A	B		A	B	
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	1	0
0	1	1	1	0	0
1	0	0	1	0	0
1	0	1	1	1	0
1	1	0	1	1	1
1	1	1	0	0	1

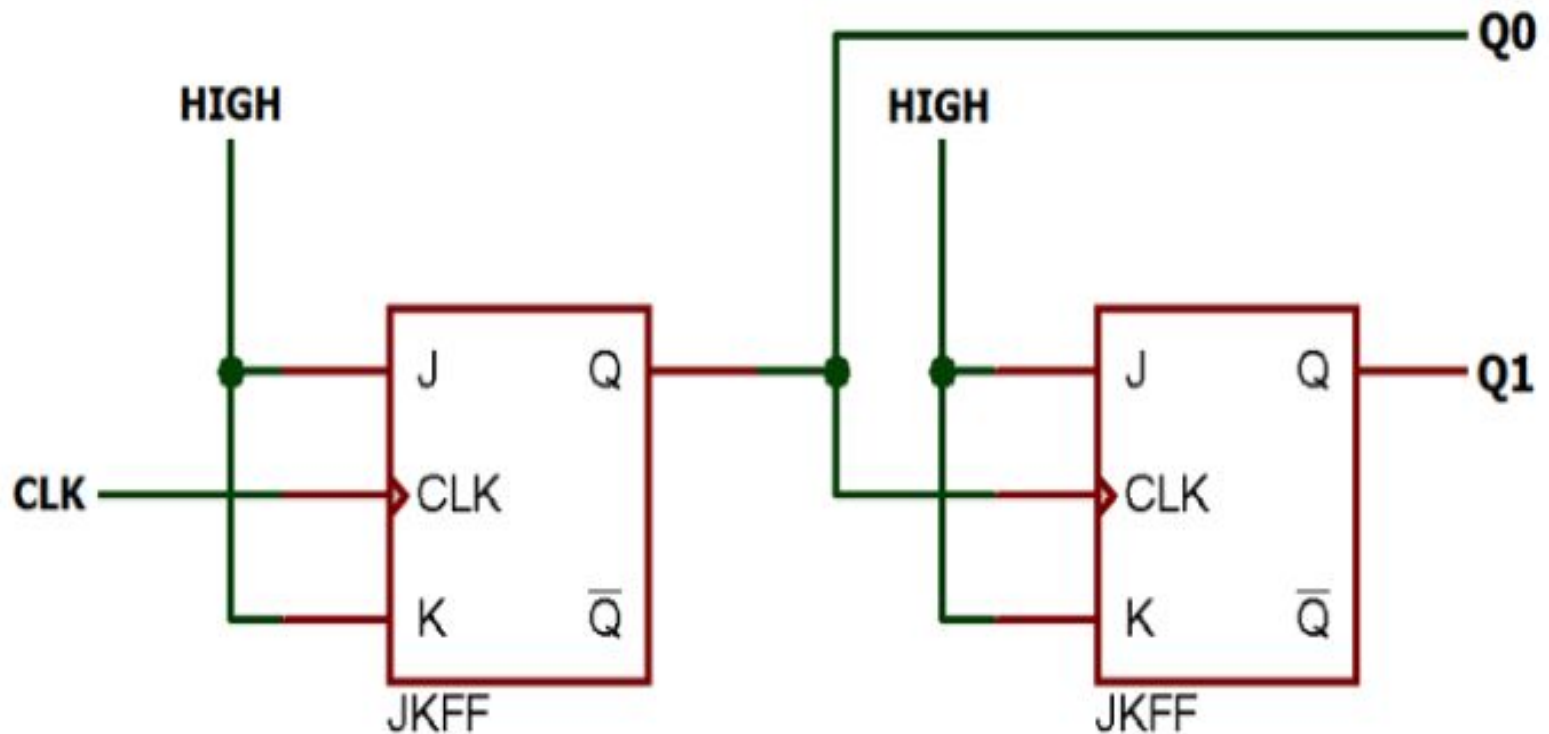
Analysis with T Flip Flop



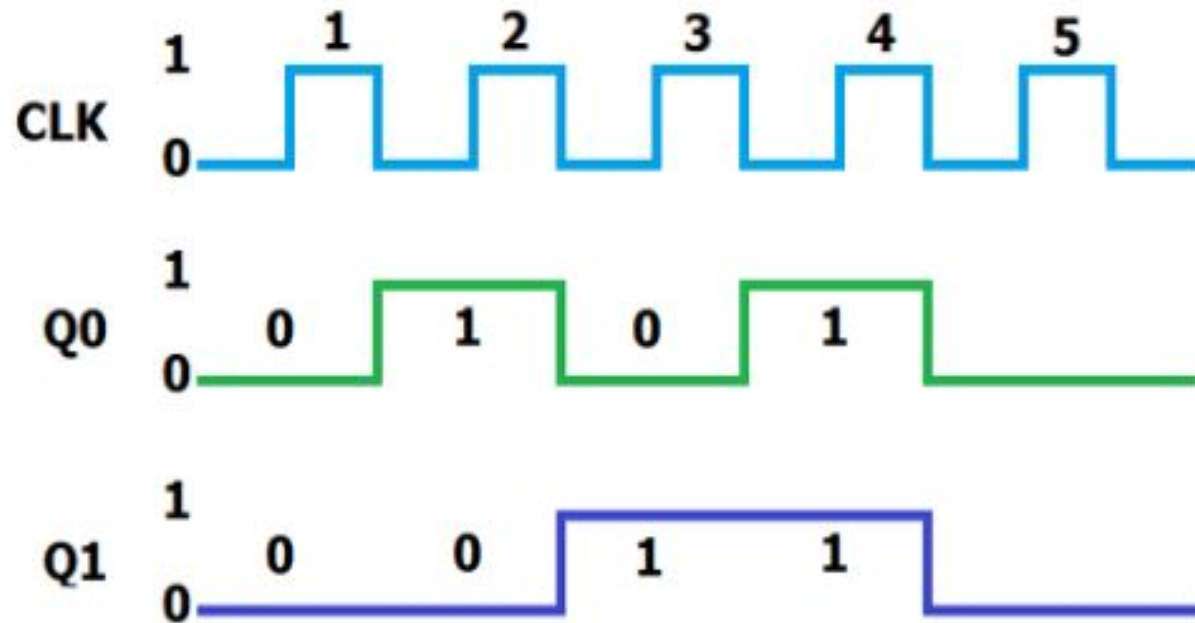
(b) State diagram

- Assignment:
- Mealy And Moore State Machines

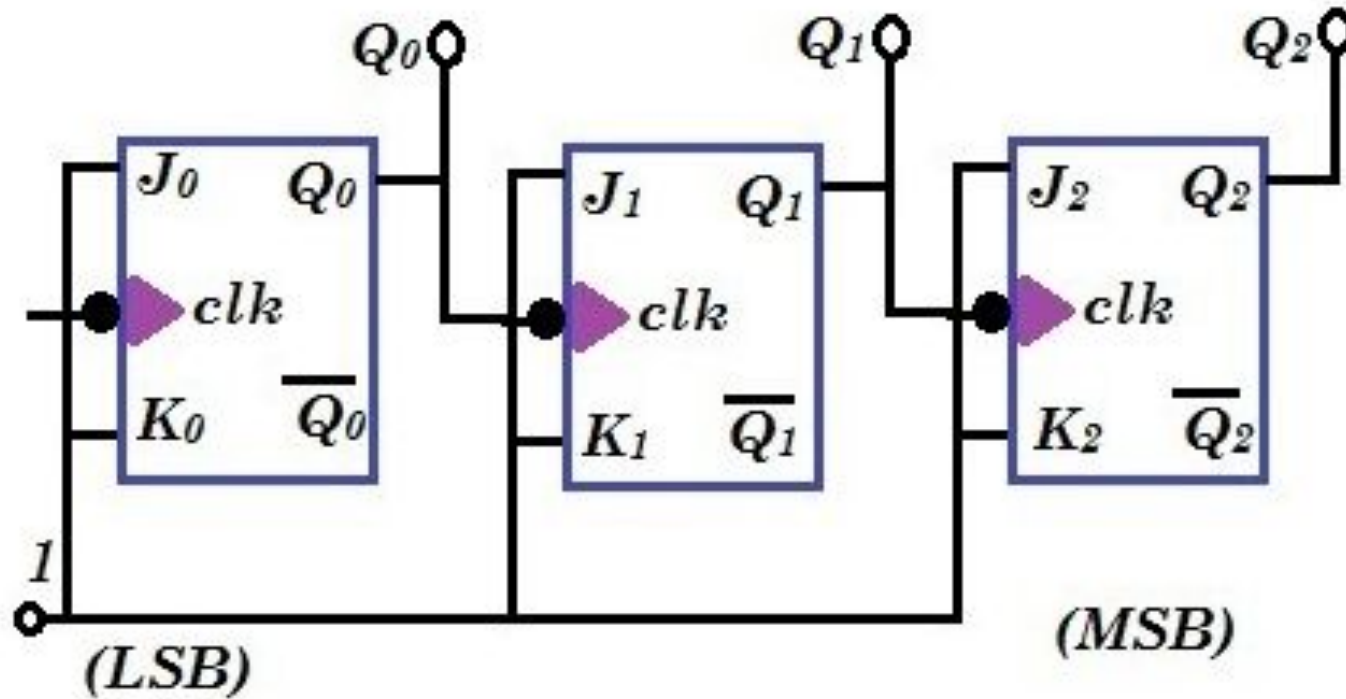
Asynchronous Counters



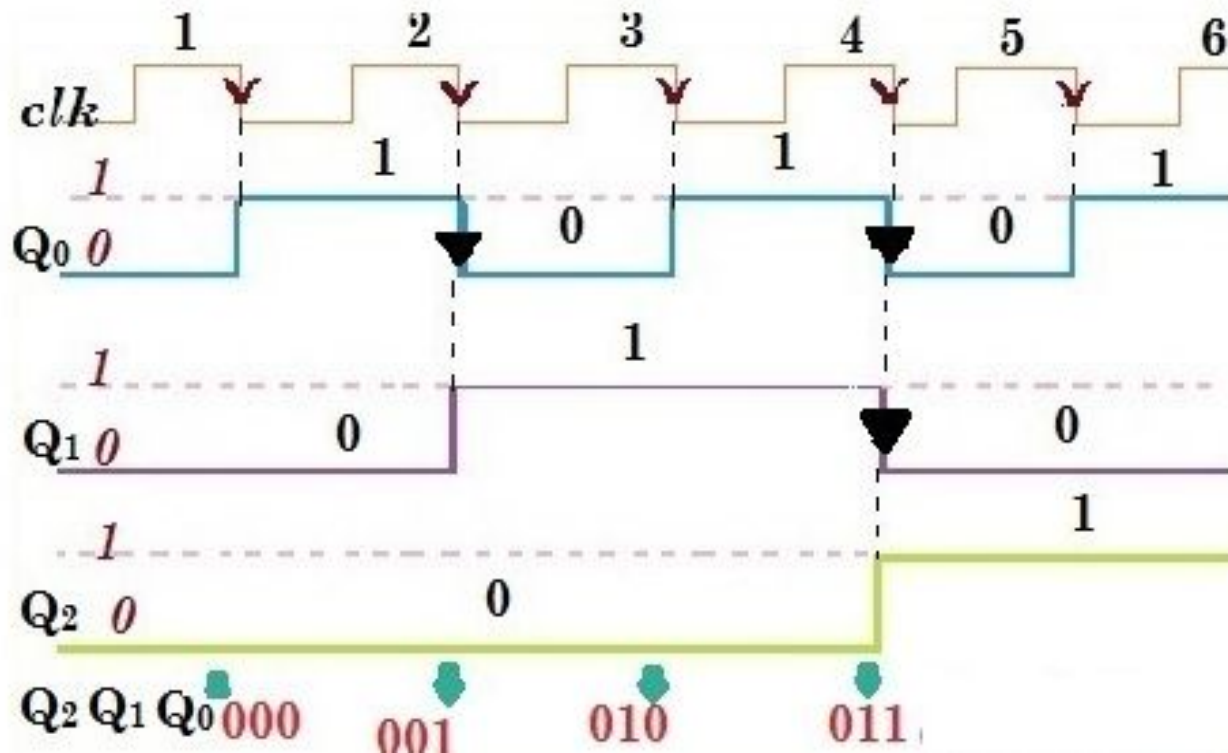
Asynchronous Counters



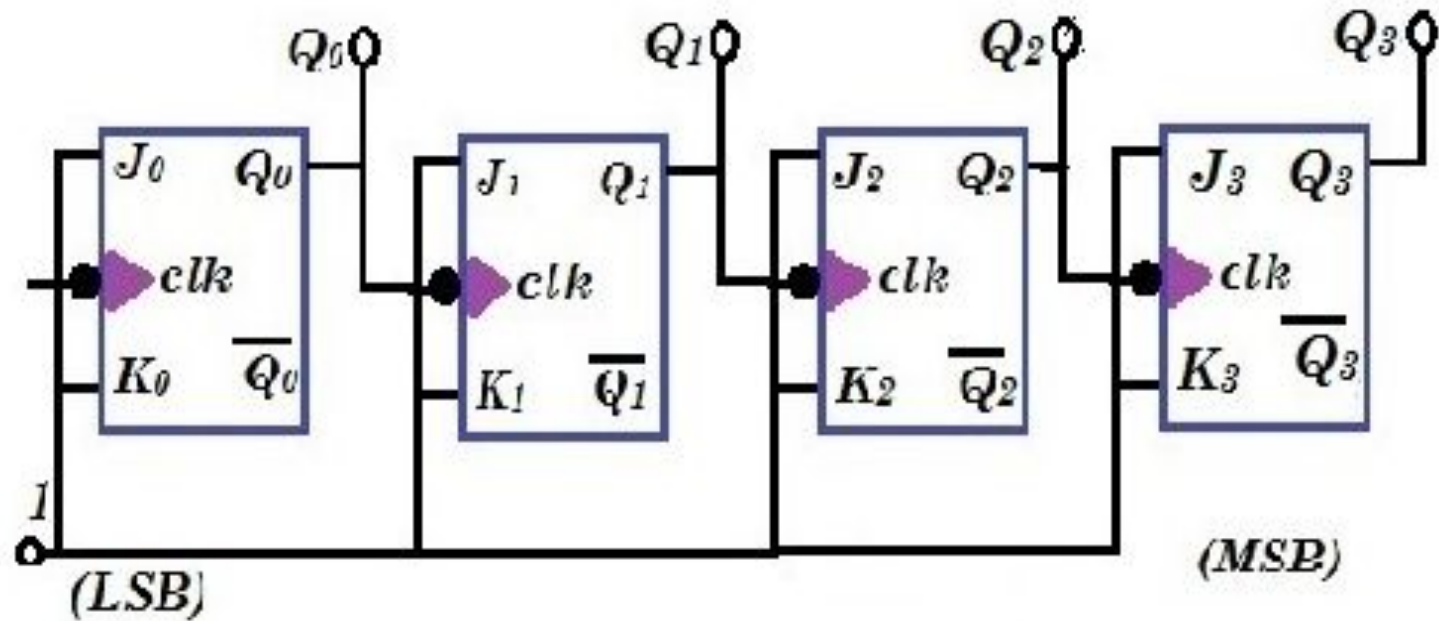
3 Bit Asynchronous Up Counter



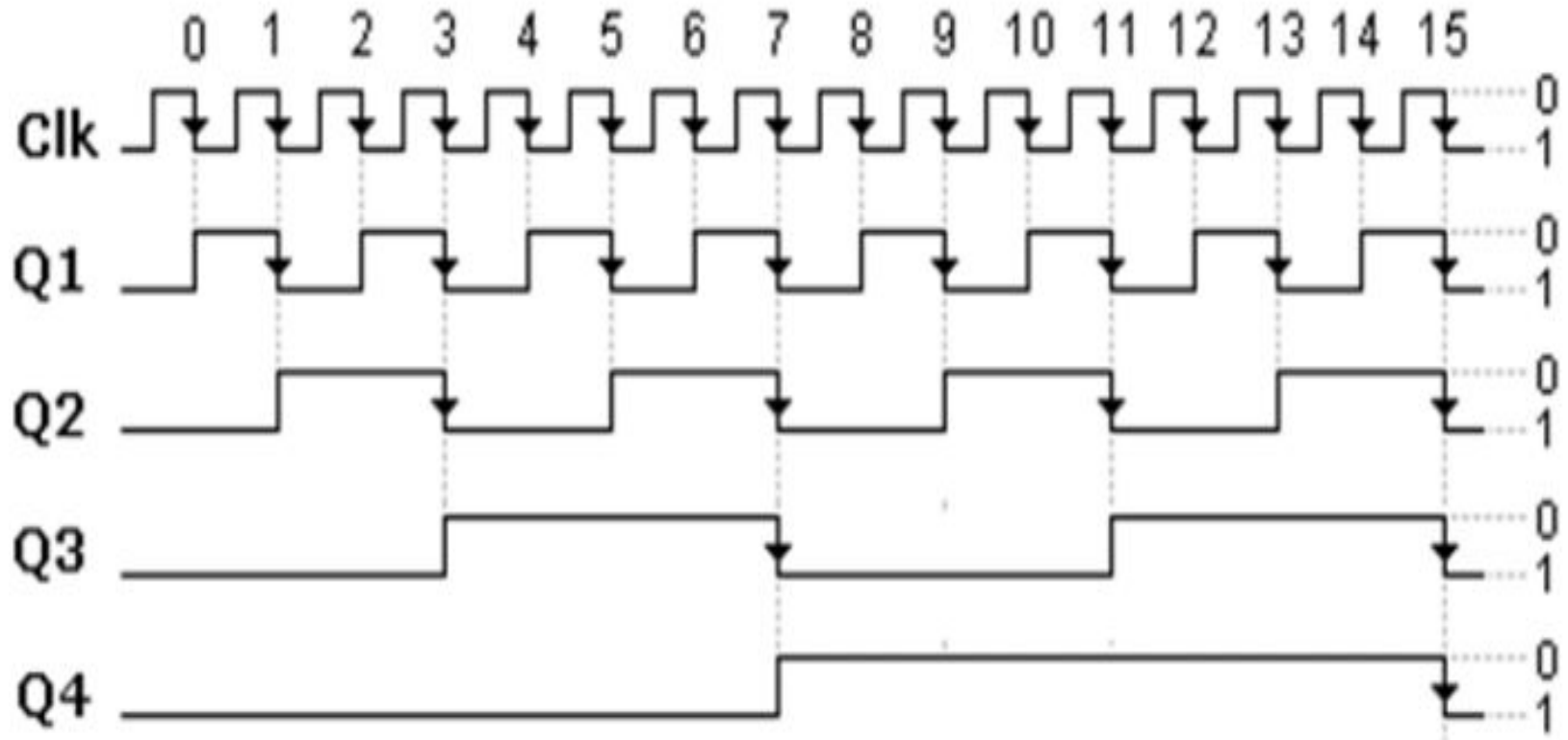
3 Bit Asynchronous Up Counter



4 Bit Asynchronous Up Counter

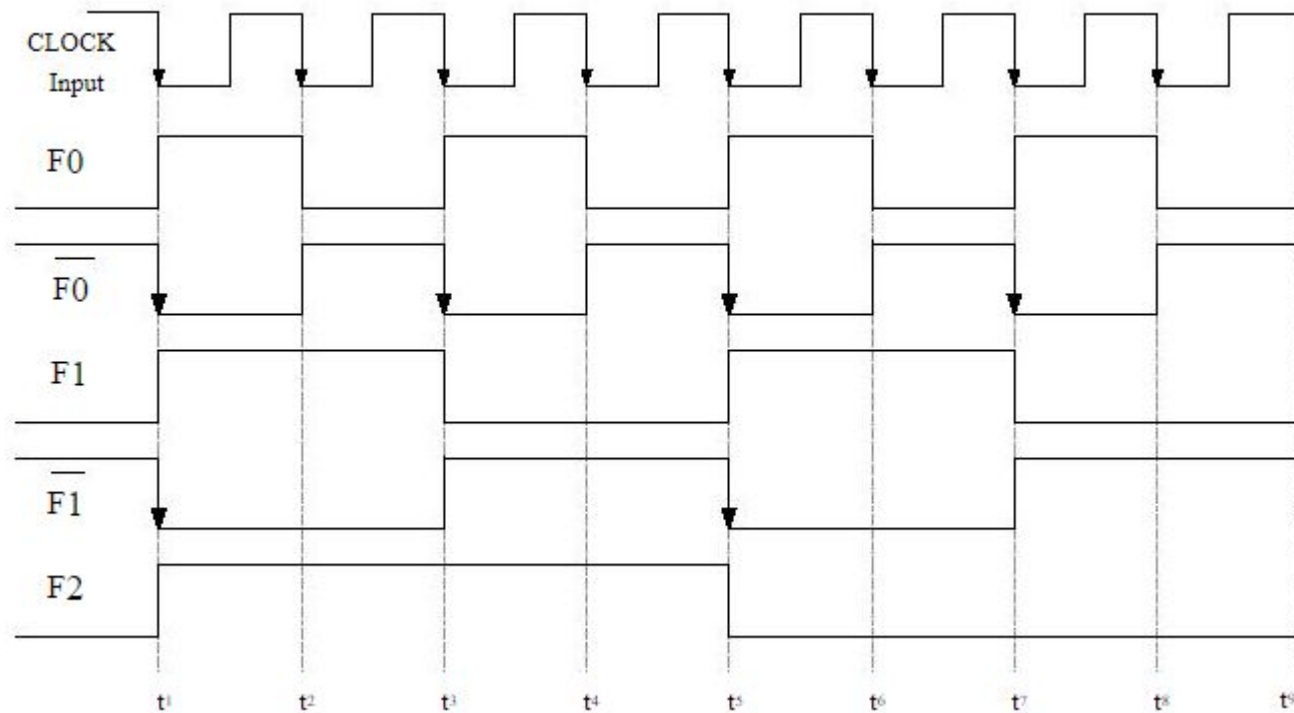


4 Bit Asynchronous Up Counter



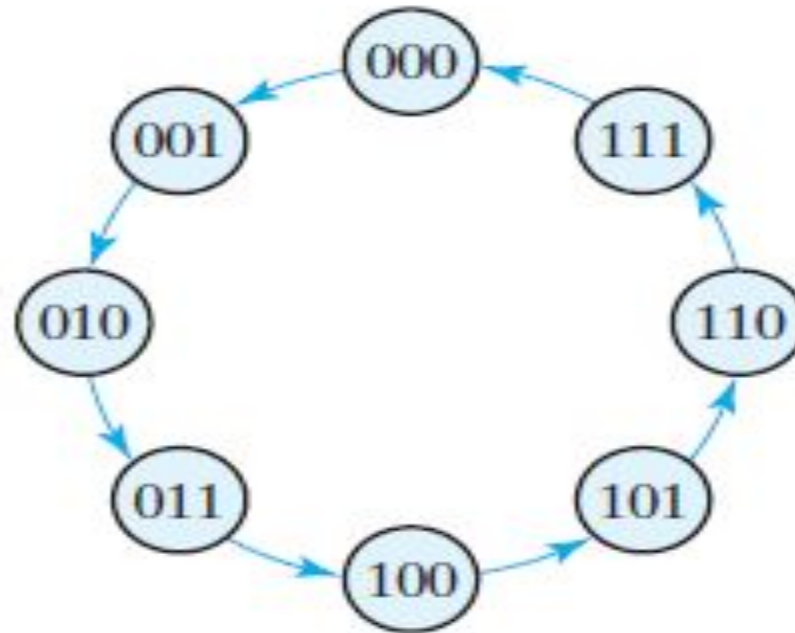
3-Bit Asynchronous Down-Counter

3-Bit Asynchronous Down-Counter



4 Bit Asynchronous Up/Down Counter

3-Bit Synchronous Up Counter



3-Bit Synchronous Up Counter

3-Bit Synchronous Up Counter

$A_2 \backslash A_1 A_0$		A_1			
		00	01	11	10
A_2	0	m_0	m_1	m_3 1	m_2
	1	m_4	m_5	m_7 1	m_6
		A_0			

$$T_{A2} = A_1 A_0$$

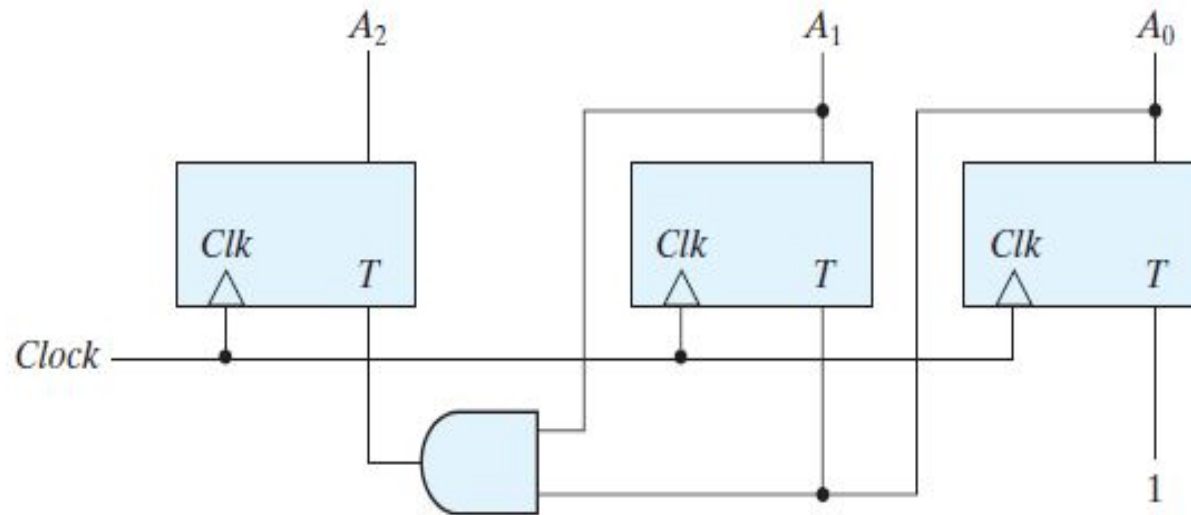
$A_2 \backslash A_1 A_0$		A_1			
		00	01	11	10
A_2	0	m_0	m_1 1	m_3 1	m_2
	1	m_4	m_5 1	m_7 1	m_6
		A_0			

$$T_{A1} = A_0$$

$A_2 \backslash A_1 A_0$		A_1			
		00	01	11	10
A_2	0	m_0 1	m_1 1	m_3 1	m_2 1
	1	m_4 1	m_5 1	m_7 1	m_6 1
		x			

$$T_{A0} = 1$$

3-Bit Synchronous Up Counter



4-Bit Synchronous Up Counter

Excitation table of JK FF

Present state Q_n	Next state Q_{n+1}	J	K
0	0	0	x
0	1	1	x
1	0	x	1
1	1	x	0

Circuit excitation table

Present state				Next state				Flip-flop input							
Q_D	Q_C	Q_B	Q_A	Q_{D+1}	Q_{C+1}	Q_{B+1}	Q_{A+1}	J_D	K_D	J_C	K_C	J_B	K_B	J_A	K_A
0	0	0	0	0	0	0	1	0	x	0	x	0	x	1	x
0	0	0	1	0	0	1	0	0	x	0	x	1	x	x	1
0	0	1	0	0	0	1	1	0	x	0	x	x	0	1	x
0	0	1	1	0	1	0	0	0	x	1	x	x	1	x	1
0	1	0	0	0	1	0	1	0	x	x	0	0	x	1	x
0	1	0	1	0	1	1	0	0	x	x	0	1	x	x	1
0	1	1	0	0	1	1	1	0	x	x	0	x	0	1	x
0	1	1	1	1	0	0	0	1	x	x	1	x	1	x	1
1	0	0	0	1	0	0	1	x	0	0	x	0	x	1	x
1	0	0	1	1	0	1	0	x	0	0	x	1	x	x	1
1	0	1	0	1	0	1	1	x	0	0	x	x	0	1	x
1	0	1	1	1	1	0	0	x	0	1	x	x	1	x	1
1	1	0	0	1	1	0	1	x	0	x	0	0	x	1	x

4-Bit Synchronous Up Counter

Present state				Next state				Flip-flop input							
Q_D	Q_C	Q_B	Q_A	Q_{D+1}	Q_{C+1}	Q_{B+1}	Q_{A+1}	J_D	K_D	J_C	K_C	J_B	K_B	J_A	K_A
1	1	0	1	1	1	1	0	x	0	x	0	1	x	x	1
1	1	1	0	1	1	1	1	x	0	x	0	x	0	1	x
1	1	1	1	0	0	0	0	x	1	x	1	x	1	x	1

For J_D

DC \ BA	00	01	11	10
00	0	0	0	0
01	0	0	1	0
11	x	x	x	x
10	x	x	x	x

$J_D = CBA$

For K_D

DC \ BA	00	01	11	10
00	x	x	x	x
01	x	x	x	x
11	0	0	1	0
10	0	0	0	0

$K_D = CBA$

For J_C

DC \ BA	00	01	11	10
00	0	0	1	0
01	x	x	x	x
11	x	x	x	x
10	0	0	1	0

$J_C = BA$

For J_B

DC \ BA	00	01	11	10
00	0	1	x	x
01	0	1	x	x
11	0	1	x	x
10	0	1	x	x

$J_B = A$

4-Bit Synchronous Up Counter

For K_C

DC \ BA	00	01	11	10
00	X	X	X	X
01	0	0	1	0
11	0	0	1	0
10	X	X	X	X

$K_C = BA$

For K_B

DC \ BA	00	01	11	10
00	X	X	1	0
01	X	X	1	0
11	X	X	1	0
10	X	X	1	0

$K_B = A$

For J_A

DC \ BA	00	01	11	10
00	1	X	X	1
01	1	X	X	1
11	1	X	X	1
10	1	X	X	1

$J_A = 1$

For K_A

DC \ BA	00	01	11	10
00	X	1	1	X
01	X	1	1	X
11	X	1	1	X
10	X	1	1	X

$K_A = 1$

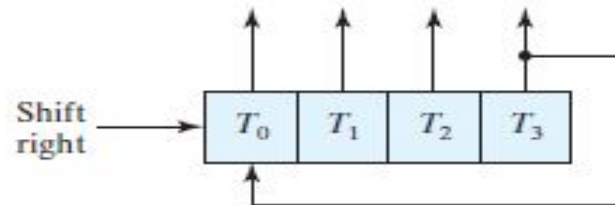
[illegible]

4-Bit Synchronous BCD Counter (0-9)

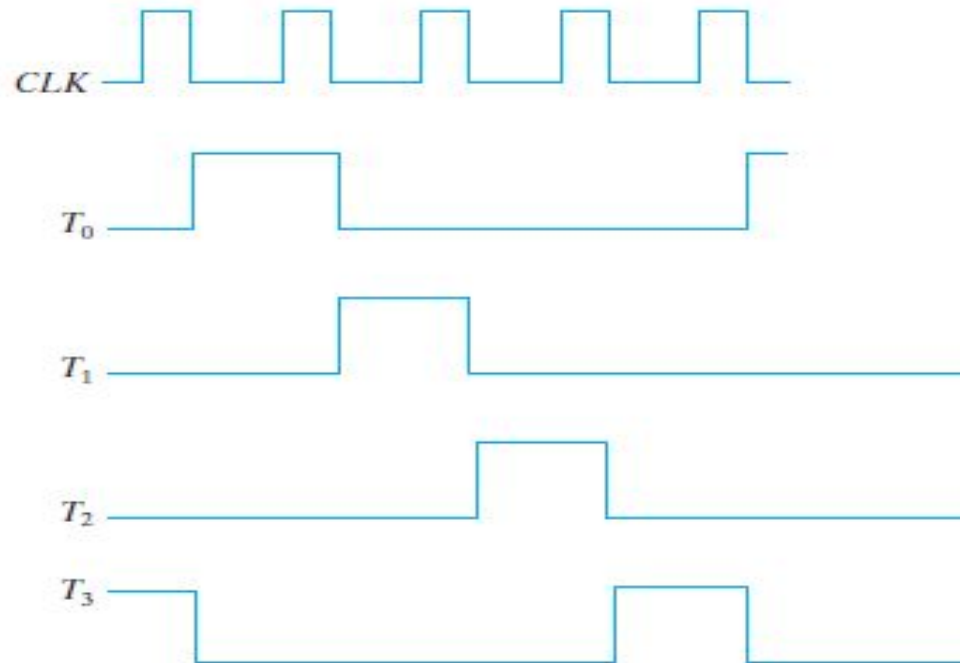
Table 6.5
State Table for BCD Counter

Present State				Next State				Output	Flip-Flop Inputs			
Q_8	Q_4	Q_2	Q_1	Q_8	Q_4	Q_2	Q_1	y	TQ_8	TQ_4	TQ_2	TQ_1
0	0	0	0	0	0	0	1	0	0	0	0	1
0	0	0	1	0	0	1	0	0	0	0	1	1
0	0	1	0	0	0	1	1	0	0	0	0	1
0	0	1	1	0	1	0	0	0	0	1	1	1
0	1	0	0	0	1	0	1	0	0	0	0	1
0	1	0	1	0	1	1	0	0	0	0	1	1
0	1	1	0	0	1	1	1	0	0	0	0	1
0	1	1	1	1	0	0	0	0	1	1	1	1
1	0	0	0	1	0	0	1	0	0	0	0	1
1	0	0	1	0	0	0	0	1	1	0	0	1

Ring Counter

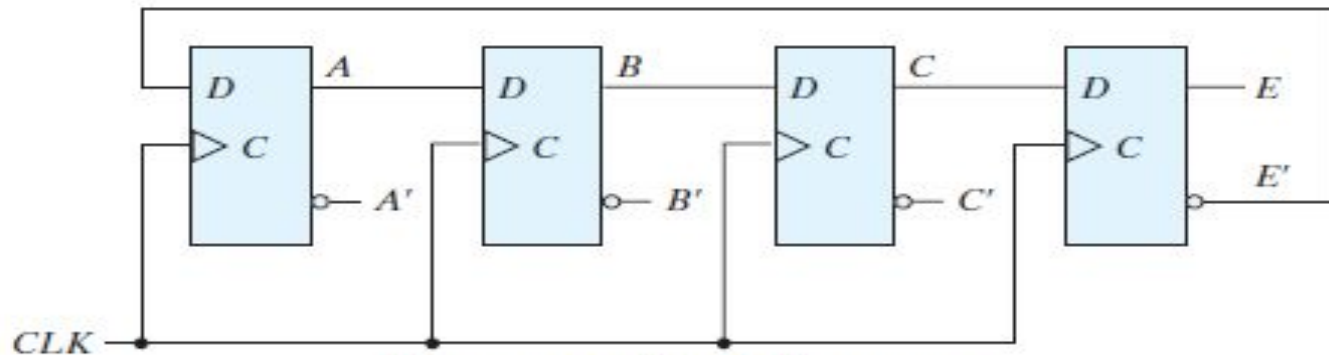


(a) Ring-counter (initial value = 1000)



(b) Sequence of four timing signals

Johnson's Counter



(a) Four-stage switch-tail ring counter

Sequence number	Flip-flop outputs				AND gate required for output
	A	B	C	E	
1	0	0	0	0	$A'E'$
2	1	0	0	0	AB'
3	1	1	0	0	BC'
4	1	1	1	0	CE'
5	1	1	1	1	AE
6	0	1	1	1	$A'B$
7	0	0	1	1	$B'C$
8	0	0	0	1	$C'E$

4-Bit Synchronous BCD Counter (0-9)

$$T_{Q1} = 1$$

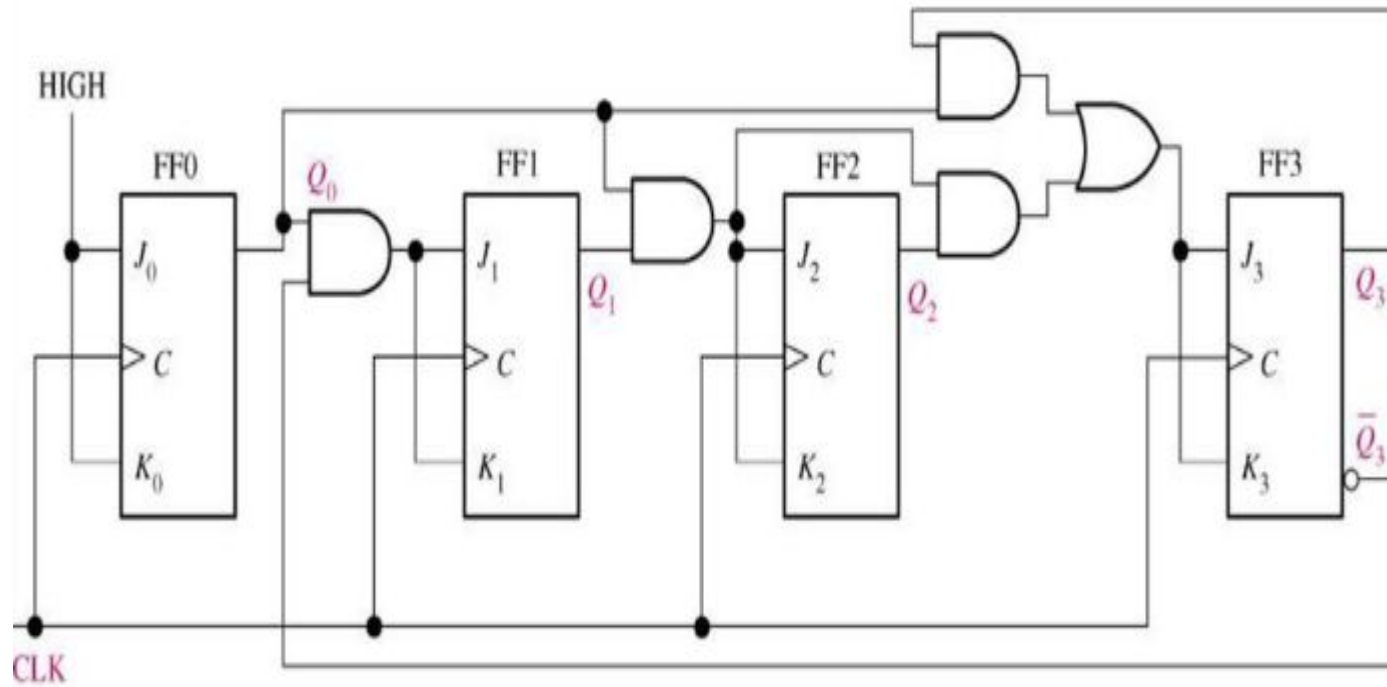
$$T_{Q2} = Q_8' Q_1$$

$$T_{Q4} = Q_2 Q_1$$

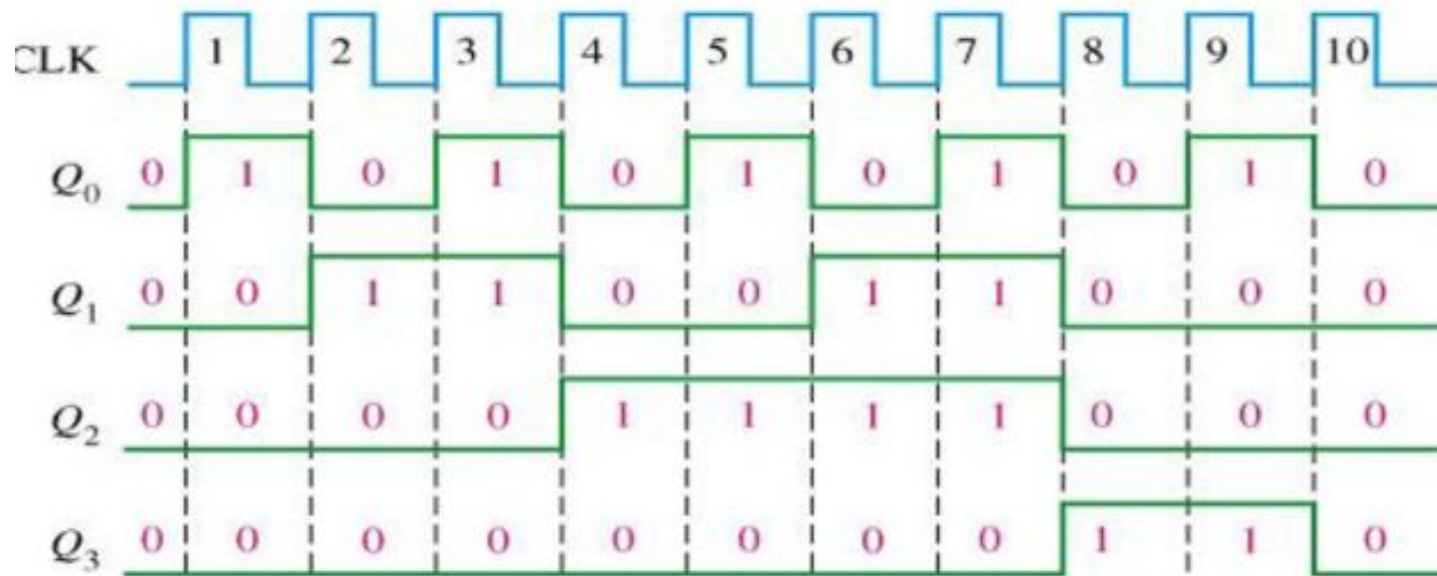
$$T_{Q8} = Q_8 Q_1 + Q_4 Q_2 Q_1$$

$$y = Q_8 Q_1$$

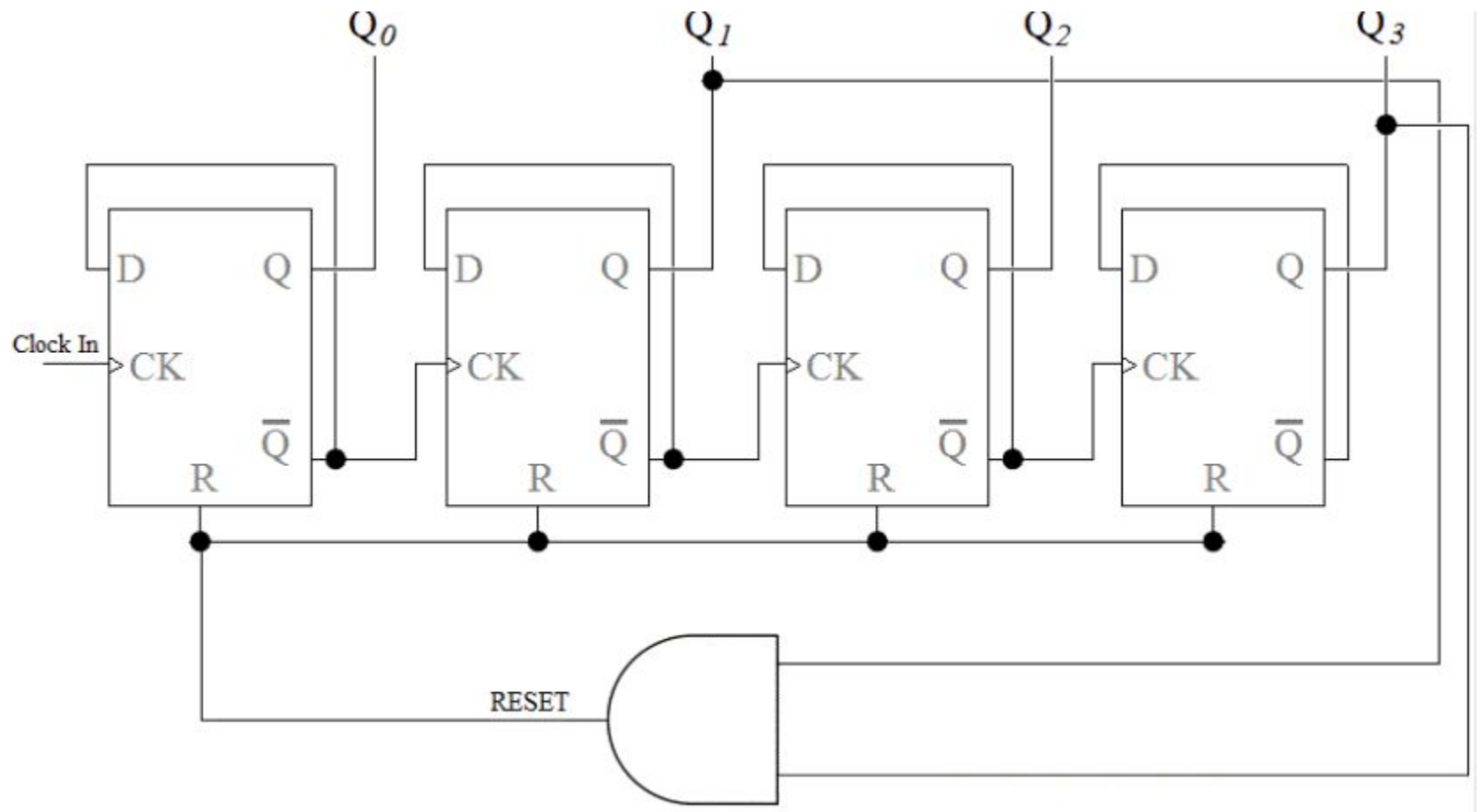
4-Bit Synchronous BCD Counter (0-9)



4-Bit Synchronous BCD Counter (0-9)



4-Bit Ripple BCD Counter (0-9)



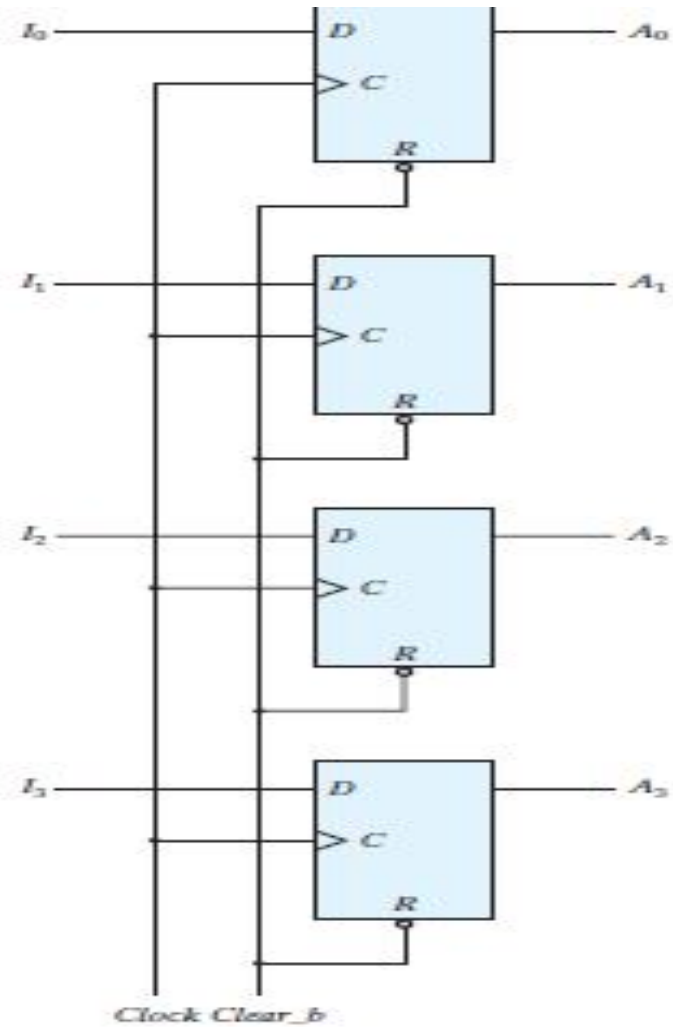
Assignment

- Binary Counter with parallel Load

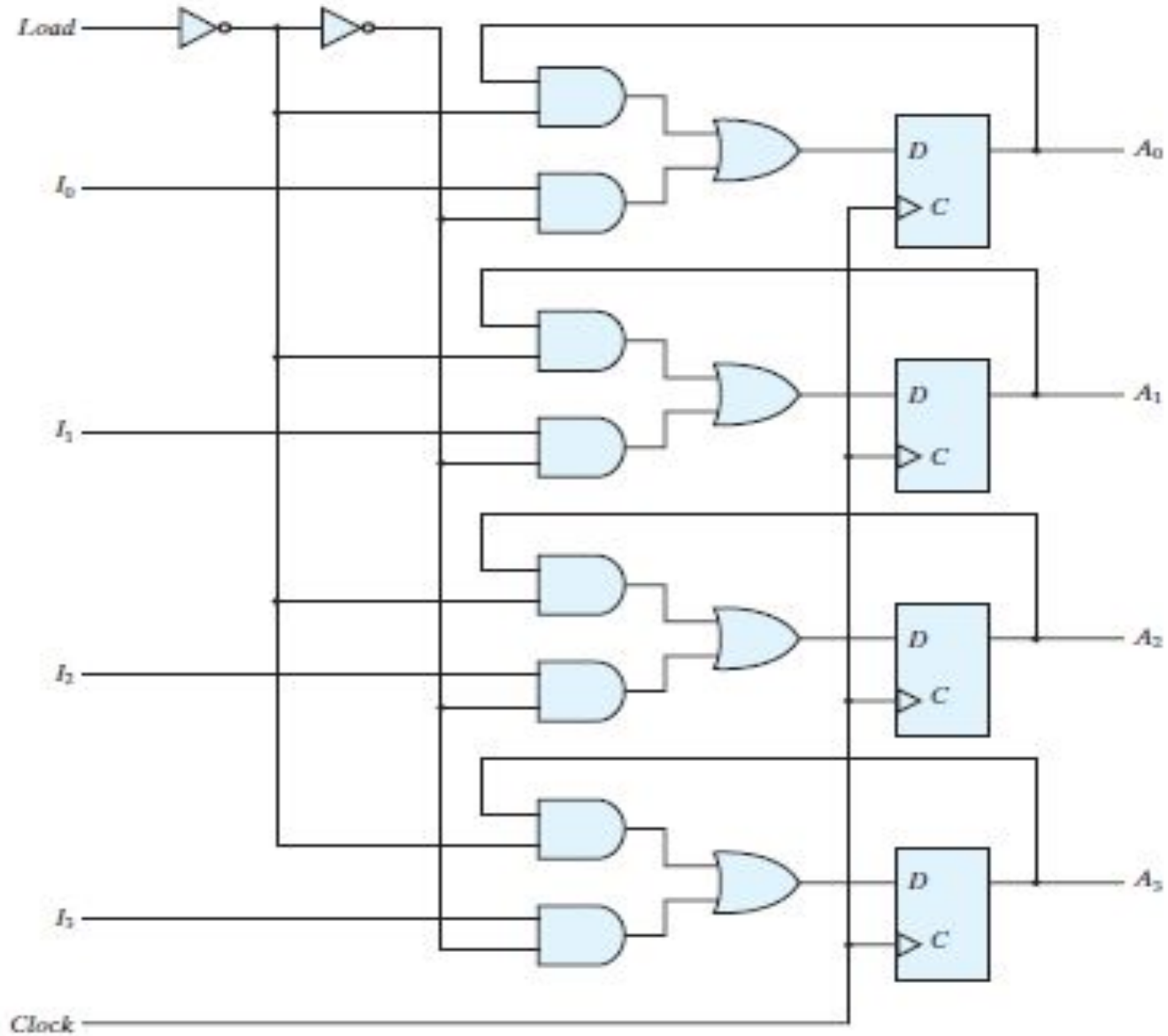
Design a 4 bit up synchronous counter

Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	x	0

Registers



Registers



Registers

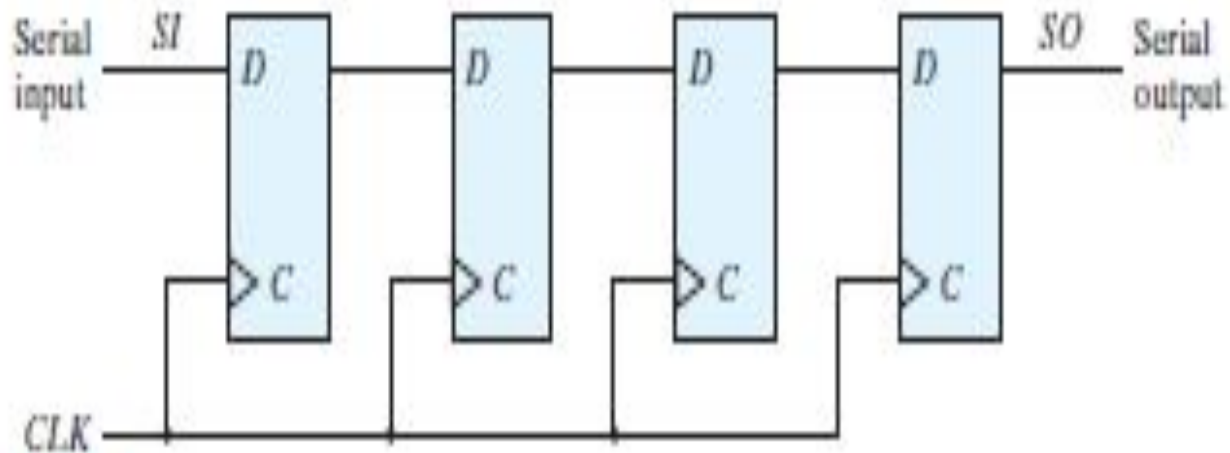
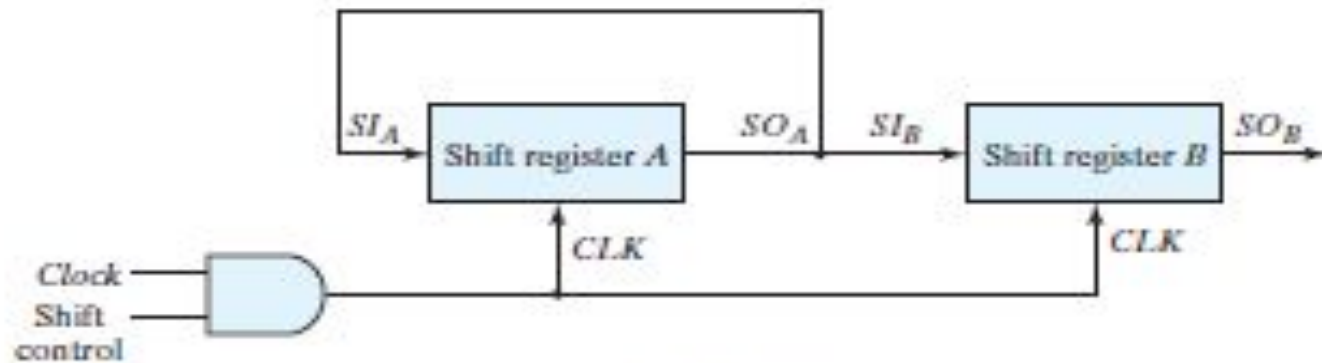
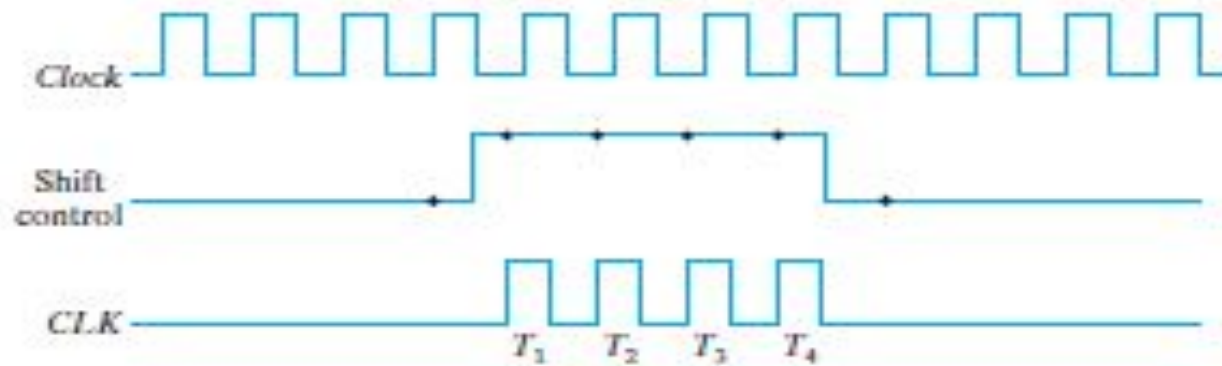


FIGURE 6.3
Four-bit shift register

Registers



(a) Block diagram



(b) Timing diagram

Registers

Table 6.1
Serial-Transfer Example

Timing Pulse	Shift Register A				Shift Register B			
Initial value	1	0	1	1	0	0	1	0
After T_1	1	1	0	1	1	0	0	1
After T_2	1	1	1	0	1	1	0	0
After T_3	0	1	1	1	0	1	1	0
After T_4	1	0	1	1	1	0	1	1

Registers

